

Explanation of Revisions

Causal Evidence for Syntax in Language Models Requires Testing Competing Explanations

We thank the Area Chair and the Reviewers of the previous cycle for their constructive and detailed reviews which helped to improve and substantially strengthen our paper. The resubmission implements the promised controls, analyses, and presentation changes. Below, we quote each reviewer comment verbatim and respond point by point. Section, figure, and table numbers refer to the revised manuscript.

Revision Summary

Issue	Revised evidence or clarification	Manuscript location
Probe validity	The same structural-probe pipeline exceeds 60% peak UUAS on UD English EWT.	Appendix C.5, Figure 13
Scalar number features	Logistic probes reach 97.79% for $\mathcal{H}_{syn}$, 98.93% for $\mathcal{H}_{prox}$, and 98.95% for $\mathcal{H}_{maj}$.	Section 6; Appendix C.6, Figure 14
Causal controls	Random-network IIA stays near 50%; trained-model source-base derangement retains high IIA across unrelated lexical pairs.	Section 7; Figures 4 and 17
DAS versus DBM	The main paper now compares rotation-based DAS and original-basis DBM directly.	Section 7, Figure 2
MAR calibration	Main results include dimension-matched null means; 1,000-sample null tests give $p < 10^{-10}$. Three-seed means and standard deviations are reported.	Section 8, Table 1; Appendix F, Tables 11–12
Dataset validation	Five linguistics PhDs annotate 100 nouns and 100 verbs. Gemini reaches 96.0% noun-animacy and 98.0% verb-category accuracy; Fleiss' κ is 0.88 and 0.92.	Section 3; Appendix B.6, Figure 7
Scope	Claims are restricted to a controlled study of synthetic English agreement. Missing template, position, OOD, and same-hypothesis baselines are stated explicitly.	Sections 1, 2, and 9; Limitations

Table 1: Main changes made for the resubmission.

Response to the Metareview

W1: Reviewers pY2P and 8woi pointed to some additional ablations and controls that would be helpful. The impressively extensive responses from the authors enacted the most important of these extensions.

Response. The proposed extensions are included as part of the revised manuscript. They include the natural-text probe sanity check, scalar number probes, dimension-matched MAR null, three-seed reporting, trained-model derangement control, subspace fractions, and Gemini validation. Detailed responses appear under [Reviewer pY2P’s probe and control responses](#) and [Reviewer 8woi’s scalar-probe and MAR responses](#).

Changes in the manuscript. Sections 3, 5–8, and 9; Figures 1–5; Appendices B.6, C.5–C.7, D, E, and F.

W2: Reviewer vfxY noted some presentational concerns about the writing.

Response. We reorganized the main paper around comparative evidence. DAS and DBM now appear in one main-text figure; the attractor, derangement, subspace-size, and MAR results are also in the main paper. Section 8 now defines how to read higher and lower MAR, gives a matched random baseline, separates MAR from model quality, and explains what it adds to IIA. The abstract, introduction, experimental setup, conclusion, and appendix were shortened and rewritten for a standalone presentation. See [Reviewer vfxY’s contribution and figure responses](#).

Changes in the manuscript. Abstract; Sections 1, 5, 7–9; Figures 2–5; Table 1; reorganized appendices.

Overall, I found the author’s responses thorough and well-done. However, the revisions suggested in these responses are extensive enough that I believe they merit another round of review so that the reviewers can fully evaluate the additions; the ARR response period is mainly intended for brief clarifications, and it therefore is not well-equipped for rigorously evaluating large-scale changes such as large numbers of additional experiments or large-scale changes to the writing. Therefore, I would encourage the authors to implement the changes proposed in their rebuttal and then resubmit that revised version.

Response. This resubmission implements those changes in the paper and appendix. The new evidence is reported with its protocol, numerical results, controls, and limitations, so it can be evaluated as part of the manuscript.

Response to Reviewer pY2P

Weaknesses

No sanity check confirming the probing pipeline can recover known structure on natural text, so it’s unclear if SYNTHAX blocks recoverability or the probe/metric is just mismatched to the setting

Response. We ran the same distance-probe objective and UUAS evaluation on the Universal Dependencies English Web Treebank. Gemma-3-12B, Qwen-3-8B, and Llama-3.1-8B each exceed 60% peak UUAS in early layers. The implementation can therefore recover dependency structure on natural text. We make the narrower inference that low SYNTHAX UUAS is specific to the controlled data and target graphs; the check does not identify which dataset property causes the difference.

Changes in the manuscript. Section 6; Appendix C.5, Figure 13.

Subspace size undermines “low-dimensional” framing: DAS subspaces exceed 1,200 dims for Gemma-3-12B; without reporting this as a fraction of hidden size, it’s unclear whether DAS is finding a compact mechanism or just fitting the mapping in near-full-rank space

Response. We removed any blanket compactness interpretation and now report both dimensions and fractions of hidden size. For example, the Gemma-3-12B syntax subspace for SG→PL spans 32.2% of its 3,840-dimensional hidden state. Singular-target interventions are smaller. We state that learned dimension is an optimization outcome, not a measure of a rule’s complexity or causal importance.

Changes in the manuscript. Section 7, Figure 5; Appendix F.2, Figure 21.

MAR lacks a chance baseline: No random-subspace null is reported, so it’s unclear whether observed MAR (0.3–0.6) reflects meaningful overlap or what dimension-counting alone would predict at these subspace sizes

Response. We added a dimension-matched Monte Carlo null. For each learned pair, we sample 1,000 random orthonormal subspace pairs with the same ambient and subspace dimensions. In the main-table settings, null means range from 0.0702 to 0.1419, while empirical means range from 0.2571 to 0.5570. Every comparison has right-tailed $p < 10^{-10}$. Thus, the observed overlap exceeds what subspace size alone predicts.

Model	Direction	MAR _{prox} (null)	MAR _{maj} (null)
Gemma-3-12B	PL→SG	0.4608 ± 0.1596 (0.0702)	0.5570 ± 0.1968 (0.1419)
	SG→PL	0.3907 ± 0.0783 (0.1211)	0.3393 ± 0.0685 (0.1186)
Qwen-3-8B	PL→SG	0.3574 ± 0.2221 (0.0895)	0.3373 ± 0.2253 (0.1113)
	SG→PL	0.2571 ± 0.0819 (0.1089)	0.4281 ± 0.2505 (0.1056)

Table 2: Revised main-paper MAR results. Values are mean ± standard deviation over three seeds; parentheses show the dimension-matched null mean.

Changes in the manuscript. Section 8, Table 1; Appendix F.3, Table 12.

Circularity risk: Because DAS directly optimizes toward each counterfactual target, multi-hypothesis high IIA could partly reflect optimizer flexibility; the untrained-model control doesn't fully rule this out (a scrambled-target control on the trained model would help)

Response. We added a trained-model derangement control. Within each target-number split, source-base pairings are scrambled so the replacement source preserves the required number but comes from an unrelated lexical context. High IIA remains, which argues against memorizing particular sentence pairs. We also retain the random-network control, where IIA stays near 50%. The manuscript now distinguishes their scope: neither control proves that the alignment is unique or that it represents the full rule rather than a transferable number feature.

Changes in the manuscript. Section 7, Figure 4; Appendix D, Figure 17.

No variance/significance reporting across seeds for IIA/UUAS/MAR

Response. We repeat the primary probe, DAS, and MAR analyses with seeds 42, 96, and 1120. Probe and intervention plots show standard-deviation bands; the appendix gives mean $\pm$ standard deviation for every reported layer. Main Table 1 reports MAR mean $\pm$ standard deviation, and the appendix gives all four models. MAR significance is evaluated against the matched Monte Carlo null.

Changes in the manuscript. Figures 1–2 and Table 1; Appendix Tables 5, 7, 8, 11, and 12.

No validation of Gemini-based lexical filtering, which is central to the dataset's confound-control claims

Response. We validate a uniform sample of 100 test nouns and 100 test verbs with five independent annotators who hold PhDs in Linguistics. Against majority labels, Gemini reaches 96.0% noun-animacy accuracy and 98.0% verb-category accuracy; Fleiss' κ is 0.88 and 0.92. WordNet checks give 98.0% noun-animacy and 95.0% broad verb-transitivity alignment. We now describe these as checks of the lexical labels, not proof that every semantic confound is absent.

Changes in the manuscript. Section 3; Appendix B.6, Figure 7.

Comments, Suggestions, and Typos

Connect MAR to existing representational similarity metrics (CKA, SVCCA, Procrustes) in related work

Response. We added this connection. SVCCA, CKA, and orthogonal Procrustes methods compare activation matrices under different invariances. MAR instead compares orthonormal bases learned by causal interventions and summarizes normalized principal-angle overlap. We also state that MAR is not a general layer- or model-similarity score.

Changes in the manuscript. Section 2, Related Work.

Add a random-subspace null distribution for MAR to justify overlap significance

Response. Added. See the response and reproduced results in Table 2 above.

Changes in the manuscript. Section 8, Table 1; Appendix F.3, Table 12.

Justify or ablate the fixed “last-token” intervention position

Response. We now state the choice precisely: interventions edit the pre-verb token, whose next-token distribution expresses the agreement decision, and evaluation compares the singular and plural forms of the same verb. Holding this position fixed makes DAS and DBM directly comparable. We did not run a position ablation and state that omission explicitly.

Changes in the manuscript. Section 5, Experimental Setup; Limitations.

Report hidden dimensionality per model to contextualize subspace sizes in Figs. 5/18

Response. Figure 5 now reports absolute size and fraction of hidden size: 3,840 for Gemma-3-12B and 4,096 for Qwen-3-8B. Appendix Figure 21 reports the same quantities for Llama-3.1-8B and OLMo-3-7B, both with hidden size 4,096.

Changes in the manuscript. Section 7, Figure 5; Appendix F.2, Figure 21.

Response to Reviewer vfxY

The dataset and probing strategies are fundamentally very incremental extensions of datasets and methods in syntactic probing that already exist. There is limited coverage of phenomena beyond S-V agreement in English and descriptions of “hierarchical syntax”, attractors and recency are quite elementary. The only distinct contribution comes in Section 8 (p8) with the MAR, but there is little room for explaining results or discussion in detail. What does it mean for models to have a higher or lower MAR? The impact of the work is, to me, unclear.

Response. We revised the paper around its distinct claim: high IIA for one hypothesis is not unique evidence when incompatible hypotheses can also be implemented, and their learned subspaces may overlap. The introduction now separates recoverability, intervention compatibility, and geometric distinctness. Section 8 defines higher MAR as greater reuse of intervention directions and lower MAR as greater separation; zero denotes orthogonality. Scores are interpreted against matched random subspaces, not as percentages or model-quality measures. The dataset and probe are presented as controlled instruments for this causal comparison, not as broad replacements for existing syntax benchmarks. Coverage remains synthetic English subject–verb agreement and is stated as such in the abstract, introduction, conclusion, and limitations.

Changes in the manuscript. Abstract; Sections 1–3, 8, and 9; Limitations.

The authors must include further experiments about whether MAR predicts OOD generalisation.

Response. MAR is not proposed as a predictor of OOD accuracy. It measures overlap between two learned intervention subspaces. We now state this distinction explicitly. Test vocabulary is disjoint, but training and test sentences share the same template families; we therefore do not present the split as OOD syntactic validation.

Changes in the manuscript. Section 2; Section 3; Limitations.

Key figures are in the appendix! “DBM fails to achieve meaningful IIA above the baseline for any of the causal hypotheses” (Fig 14) is an interesting result but currently buried in the appendix. Fig 17, Fig 18 and Table 7 are interesting results that should form the main part of the paper, but are completely underdiscussed.

Response. We moved the comparative evidence into the main paper. Revised Figure 2 places DAS and DBM layer-wise results together for Gemma and Qwen. Revised Figures 3–5 show attractor-count behavior, the trained-model derangement control, and subspace dimensions. Revised Table 1 reports Gemma and Qwen MAR with random baselines and seed variation. Section 7 now explains the DAS–DBM contrast: the same pairs and targets are accessible after rotation but not through a sparse exchange of original-basis coordinates. Section 8 explains the MAR patterns. Extended Llama and OLMo results remain in the appendix.

Changes in the manuscript. Sections 7–8; Figures 2–5; Table 1.

Authors should refer to all figures in the appendix in the main body.

Response. Every appendix figure is now cited from the main paper at the point where its corresponding method, control, or extended-model result is discussed.

Changes in the manuscript. Sections 3, 5–8, with explicit references to Appendix Figures 6–21.

What is the accuracy of Gemini’s animacy classification? Authors should try to report results for a sample in the Appendix.

Response. On a uniform sample of 100 nouns and 100 verbs, Gemini reaches 96.0% accuracy for noun animacy and 98.0% for verb categorization against the five-expert majority label. Fleiss’ κ is 0.88 and 0.92. The appendix also reports 98.0% noun-animacy alignment with WordNet and shows the annotation interface.

Changes in the manuscript. Section 3; Appendix B.6, Figure 7.

Comments, Suggestions, and Typos

Authors can do a much better job of summarising key results in figures in the main body – too much space is wasted on figures that are not comparative; this is exactly where the contribution of the paper lies.

Response. The main paper now prioritizes comparisons. Figure 2 directly contrasts DAS and DBM; Figure 3 compares hypotheses across attractor counts; Figure 4 shows the derangement control; Figure 5 compares learned dimensions; and Table 1 compares MAR with chance. Non-central extended-model plots and numerical tables remain in the appendix.

Changes in the manuscript. Sections 6–8; Figures 1–5; Table 1.

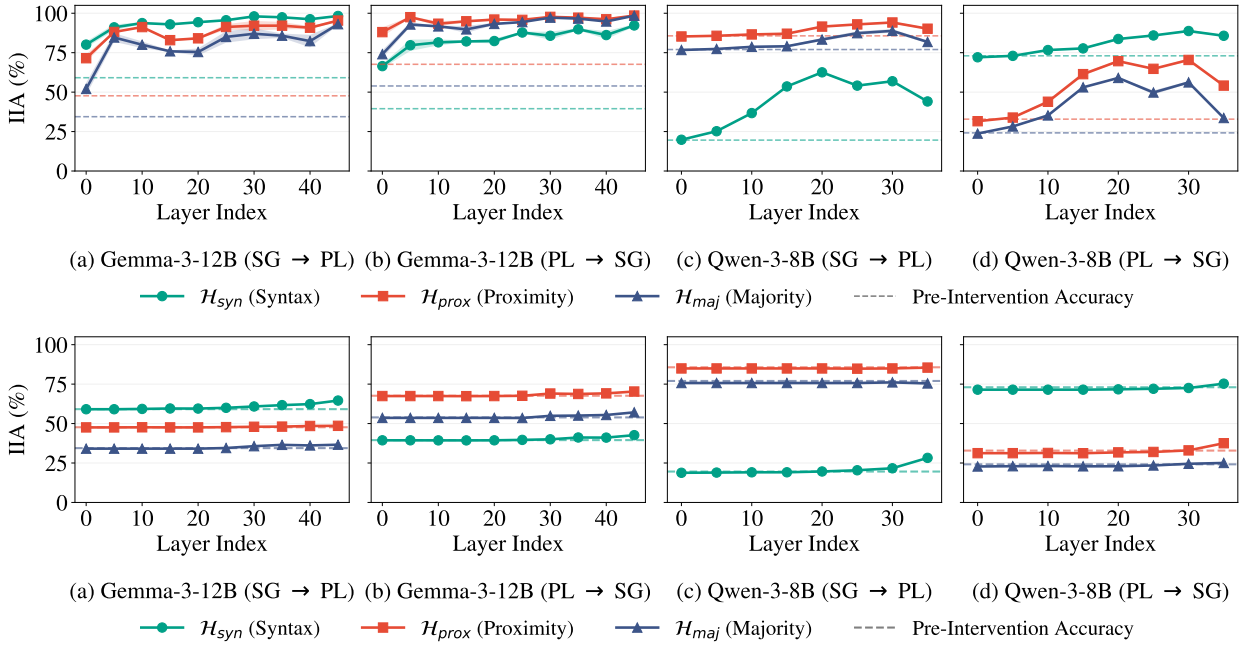


Figure 1: Revised main-paper Figure 2. Top: Boundless DAS. Bottom: DBM on the same 4-attractor intervention targets. Dashed lines are pre-intervention baselines.

Response to Reviewer 8woi

The strongest claim is broader than the evidence. The results show that, in SYNTHAX subject-verb agreement, Boundless DAS can align multiple counterfactual hypotheses with high interchange intervention accuracy. This supports caution against single-hypothesis causal claims, but it does not establish that causal evidence for syntax generally requires this exact competing-explanations framework or that prior syntax-circuit claims are invalid.

Response. We narrowed the claim. The paper now describes a controlled case study of three agreement rules and explicitly says that it neither shows that LMs lack syntax nor invalidates prior circuit findings. The supported methodological conclusion is narrower: high IIA makes a hypothesis intervention-compatible, but does not uniquely identify it when competing hypotheses also achieve high IIA. The abstract, introduction, conclusion, and limitations now use this scope consistently.

Changes in the manuscript. Abstract; Sections 1 and 9; Limitations.

The synthetic setting may introduce artifacts that are hard to separate from the intended syntax/heuristic distinction. SYNTHAX controls many confounds, but template-generated relative clauses and repeated attractor patterns may create unnatural distributions. Models may rely on template-specific or position-specific features rather than general syntactic mechanisms. A held-out-template split or semi-naturalistic validation set is important for the central conclusion.

Response. SYNTHAX provides a controlled setting for comparing syntax with competing heuristics while limiting lexical confounds. The revision clarifies that its two template families are shared across splits, so the results establish causal implementability within this grammar rather than held-out-template generalization. A natural-text probe check supports transfer beyond synthetic sentences at the probe level. We identify held-out templates and semi-natural causal evaluation as important next steps in the Limitations section.

Changes in the manuscript. Sections 1 and 3; Limitations.

Comments, Suggestions, and Typos

Add simple linear/classifier probes for the three scalar number features: structural subject number, nearest-noun number, and majority number. This would help interpret the low UUAS result.

Response. We added logistic-regression probes on Gemma-3-12B’s final-token representations at layer 45 in the 4-attractor split. Test accuracy is 97.79% for $\mathcal{H}_{syn}$, 98.93% for $\mathcal{H}_{prox}$, and 98.95% for $\mathcal{H}_{maj}$. Thus, the three scalar features are linearly accessible even though the distance probe does not recover their target graphs. We avoid the stronger inference that classifier accuracy alone proves causal use or geometric entanglement.

Changes in the manuscript. Section 6; Appendix C.6, Figure 14.

Provide random-subspace and same-hypothesis-retraining baselines for MAR, with confidence intervals over DAS random seeds.

Response. We added the dimension-matched random-subspace baseline and repeat DAS/MAR with seeds 42, 96, and 1120. Main Table 1 reports mean $\pm$ standard deviation and the matched null mean; Appendix Table 12 gives 1,000-sample null tests, and Appendix Table 11 gives all four models. We did not run a same-hypothesis retraining baseline. The Limitations section now states that this baseline would better separate cross-hypothesis differences from optimization variation even when the target explanation is unchanged.

Changes in the manuscript. Section 8, Table 1; Appendix F, Tables 11–12; Limitations.

Limitations and Societal Impact

The limitations section is appropriate and acknowledges the synthetic scope, the possibility of unmeasured heuristics, linear-intervention constraints, and model-scale limitations. The societal impact discussion is also reasonable for an interpretability paper. I would add that interpretability claims based on insufficiently tested causal mechanisms could mislead downstream safety evaluations, while the proposed competing-explanations protocol may reduce this risk by making mechanistic claims more falsifiable.

Response. We added this point directly. The Ethical Considerations section now states that an insufficiently tested causal claim could mislead downstream safety evaluation, and that testing competing explanations and reporting non-unique alignments can make such claims easier to challenge and falsify.

Changes in the manuscript. Ethical Considerations.

Causal Evidence for Syntax in Language Models Requires Testing Competing Explanations

Anonymous ACL submission

Abstract

Autoregressive language models (LMs) often appear to track hierarchical syntax, yet it remains unclear whether this behavior reflects genuine structural representation or the exploitation of shallow statistical heuristics. We introduce **SYNTHAX**, a synthetic testbed that isolates the causal effects of hierarchical syntax, proximity, and majority-number heuristics by systematically varying attractor configurations to prescribe conflicting counterfactual targets. Applying structural probes, we find that they fail to recover hypothesis-specific agreement structure; however, using Boundless Distributed Alignment Search (DAS), we demonstrate that models can be steered toward these competing hypotheses. Differential Binary Masking remains near the pre-intervention baseline, showing that the effective DAS directions are not sparse coordinates in the original activation basis. To measure the geometric overlap between these learned intervention subspaces, we introduce the **Manifold Alignment Ratio (MAR)**. Syntax and heuristic interventions often reuse directions, and their overlap exceeds dimension-matched random-subspace baselines. We conclude that causal interpretability requires evaluating proposed mechanisms against plausible alternatives and comparing their geometric structure, rather than treating individual intervention success as a definitive explanation.¹

1 Introduction

Though trained solely on a next-token prediction objective, autoregressive language models (LMs) capture surface regularities and local n -grams (Svete and Cotterell, 2024; Nguyen, 2024). This behavior frequently motivates the critique that LMs operate, at least in part, as “stochastic parrots” (Bender et al., 2021). In subject–verb agreement,

the verb must match the structural subject regardless of intervening attractor nouns. Consequently, when a model predicts the correct verb, it might be tracking genuine hierarchical structure, but it could also be exploiting shallower, correlated heuristic cues.

Existing targeted syntactic evaluations, such as BLiMP (Warstadt et al., 2020) and SyntaxGym (Gauthier et al., 2020), make this linguistic competence measurable by testing whether LMs assign higher probabilities to grammatical sentences than to ungrammatical ones. While these behavioral benchmarks show sensitivity to long-distance dependencies, they do not reveal the mechanism driving a prediction. Behavioral success is therefore ambiguous: a correct agreement prediction is compatible with tracking the structural subject, copying the nearest noun, or defaulting to the majority number among nouns.

These evaluations often rely on a narrow set of lexical items, which can obscure whether a model is truly systematic or simply exploiting frequency confounds (Newman et al., 2021). This motivates a setting in which the vocabulary and the competing agreement cues can both be controlled.

Interpretability research has traditionally used structural probing to test whether syntactic trees are linearly recoverable from hidden states (Hewitt and Manning, 2019; White et al., 2021; Müller-Eberstein et al., 2022; Simon et al., 2024, 2025). Recoverability, however, does not establish causal use. Causal abstraction methods such as Distributed Alignment Search (DAS) (Geiger et al., 2024) and its boundless variant (Wu et al., 2023) instead isolate subspaces that causally steer model behavior. Yet CausalGym shows that expressive alignment methods can also succeed on control tasks and shallow heuristics (Arora et al., 2024). Consequently, testing only a hierarchical explanation is insufficient; a successful intervention may appear to confirm syntax even when competing

¹Our source code and dataset are attached as supplementary material and will be released upon acceptance.

heuristics can steer the same prediction.

To investigate this ambiguity, we compare competing causal explanations side by side using subject–verb agreement. The phenomenon admits three precise, competing rules: hierarchical syntax agrees with the structural subject, proximity agrees with the closest noun, and majority number agrees with the most frequent plurality among intervening nouns. Since these mechanisms can prescribe distinct counterfactual verb pluralities for the same input, they can be compared under an identical intervention protocol. This design separates three questions: whether each structure is recoverable, whether each rule can causally steer a prediction, and whether successful interventions occupy distinct subspaces.

To answer these questions, we introduce SYNTHAX, a controlled synthetic testbed for causal interpretation of subject–verb agreement. It uses a controlled vocabulary and systematically varies the number and type of attractors to reduce lexical and semantic frequency confounds. Our goal is not to replace broad-coverage benchmarks such as SyntaxGym, but to provide a complementary environment where competing causal models can be probed, intervened on, and geometrically compared. We evaluate Gemma-3-12B, Qwen-3-8B, Llama-3.1-8B, and OLMo-3-7B. The testbed isolates synthetic English agreement; it does not stand in for syntax as a whole.

Our main contributions are:

- **A causal testbed for syntax.** We introduce SYNTHAX (§3), a synthetic subject–verb agreement dataset designed to reduce distributional shortcuts and systematically vary attractor configurations for causal interpretability. Expert annotation and WordNet checks validate its semantic filtering.
- **Conflicting causal models for the same behavior.** We formalize hierarchical syntax, proximity, and majority number as competing causal explanations and evaluate them under an identical intervention protocol. Structural probes fail to recover meaningful hypothesis-specific structure (§6), while Boundless DAS finds effective intervention subspaces for both hierarchical and heuristic explanations (§7). DBM remains near the corresponding baselines, providing an original-basis control for the rotated DAS search.
- **A geometric check on intervention evidence.** We introduce the Manifold Alignment Ratio (MAR) to compare learned intervention subspaces

across causal models (§8). The results show that conflicting high-intervention explanations can share representation space within the residual stream. Dimension-matched random baselines and three optimization seeds calibrate this overlap.

Together, these findings separate what is recoverable from a representation, what can causally steer a model, and whether competing causal explanations are geometrically distinct. We do not conclude that LMs lack syntactic representations or that causal interventions are uninformative. Rather, causal syntax interpretation requires a stronger standard of proof: proposed mechanisms should be tested against plausible alternatives and compared geometrically before being accepted as the definitive explanation of model behavior.

2 Related Work

Extensive behavioral work shows that recurrent and Transformer-based LMs track subject–verb agreement but often falter when structural and linear cues conflict (Linzen et al., 2016; Gulordava et al., 2018; Marvin and Linzen, 2018). Structural probing tests whether syntactic information is recoverable from hidden representations (Adi et al., 2017; Conneau et al., 2018; Hewitt and Manning, 2019; Tenney et al., 2019; Simon et al., 2024), but recoverability does not establish causal necessity (Hewitt and Liang, 2019; Pimentel et al., 2020; Elazar et al., 2021). Interventional methods address this gap by editing activations to test whether behavioral changes match specific causal hypotheses (Geiger et al., 2021, 2024; Wu et al., 2023). CausalGym shows that high causal efficacy does not guarantee a faithful or unique mechanism, because expressive methods can also align control features or shallow heuristics (Arora et al., 2024). We make this ambiguity explicit by comparing structural and heuristic explanations under one protocol.

MAR is related to representational similarity measures such as SVCCA, centered kernel alignment (CKA), and orthogonal Procrustes comparisons (Raghu et al., 2017; Kornblith et al., 2019; Ding et al., 2021). Those methods compare activation matrices over examples, allowing different invariances and alignments. MAR instead compares two orthonormal bases already learned by causal interventions and summarizes their principal-angle overlap, normalized for unequal subspace dimensions. It therefore answers a narrower question:

how much do two intervention-compatible explanations reuse projection directions? MAR is not a general similarity score between layers or models, and our work does not establish it as a predictor of out-of-distribution accuracy.

3 Dataset Construction: SYNTHAX

While broad-coverage benchmarks effectively evaluate general linguistic competence, they do not disentangle specific, competing processing mechanisms. To systematically evaluate these mechanisms without distributional shortcuts, we introduce SYNTHAX, a fully synthetic, template-based generation pipeline. It adapts recursive relative-clause templates from [Marvin and Linzen \(2018\)](#) and varies the number and number feature of intervening nouns. Its purpose is to create configurations in which hierarchical and surface cues prescribe different counterfactual targets.

3.1 Lexicon, Templates, and Splits

Our structural templates rely on a controlled vocabulary to prevent semantic anomalies and n -gram shortcuts, constructed via a three-stage filtering pipeline, as illustrated in Figure 6.

Morphological Inflection. We use UniMorph to acquire English morphological paradigms. This provides a deterministic lookup table for the required plural nouns and verb conjugations.

Frequency Filtering. We filter the lexicon with wordfreq, discarding any item with a Zipf score below 3.5. This removes archaic and domain-specific vocabulary that could make token familiarity an avoidable confound.

Semantic Categorization. A central challenge in synthetic generation is avoiding selectional violations: grammatically well-formed but implausible sentences such as “The idea eats.” We use Gemini to classify noun animacy and match verbs with appropriate subjects ([Comanici et al., 2025](#)). Figures 8, 9, 10, and 11 reproduce the four filtering prompts. On a blind sample of 100 nouns and 100 verbs annotated by five annotators, Gemini agrees with the majority label on 96.0% of noun-animacy cases and 98.0% of verb-category cases; Fleiss’ κ is 0.88 and 0.92, respectively. Figure 7 shows the annotation interface, and Appendix B.6 gives the full human and WordNet checks.

With our controlled lexicon established, we adapt the recursive templates of [Marvin and Linzen](#)

(2018) to target agreement across relative clauses. The templates embed hierarchical distractors (attractors) between the main subject and the target verb. By manipulating the morphological number of these attractors, we create adversarial configurations where structural and linear cues clash. Each template contains one to four attractors. We generate 19,000 training, 1,000 validation, and 3,000 test instances. Test nouns and verbs are disjoint from the training and validation vocabularies, so evaluation requires lexical generalization within the same template family. Template components and representative counterfactual sentences appear in Tables 2 and 3.

The lexical split blocks direct reuse of the noun-verb pairs seen while fitting probes or interventions. Within each sentence, nouns and verbs are also non-repeating, and the evaluated target verbs form one token under every model tokenizer. These restrictions remove several simple memorization routes, but they do not constitute a held-out-template test: train and test items share the same two recursive template families. We therefore use the dataset to compare explanations under controlled conflicts, not to claim broad syntactic generalization.

4 Diagnostic Configurations and Causal Hypotheses

For a sentence prefix containing a main subject and k intervening nouns, we compare three deterministic rules (formalized in Appendix A):

- (i) **Hierarchical syntax** ($\mathcal{H}_{syn}$): choose the verb number of the structural main subject.
- (ii) **Proximity** ($\mathcal{H}_{prox}$): choose the verb number of the closest preceding noun.
- (iii) **Majority number** ($\mathcal{H}_{maj}$): choose the number held by most intervening attractor nouns.

For example, in “The athlete that the coach trains whom the fans admire that the sponsors fund __,” $\mathcal{H}_{syn}$ selects singular from *athlete*; $\mathcal{H}_{prox}$ selects plural from the closest noun, *sponsors*; and $\mathcal{H}_{maj}$ selects plural because two of the three attractors are plural. The main subject is excluded from the majority count, making this surface rule distinct from hierarchical agreement.

4.1 Intervention Directions and Data Splits

We evaluate two directions. In **SG**→**PL**, the base sentence has a singular main subject: the counterfactual target is plural for $\mathcal{H}_{syn}$ and singular for the two heuristic splits. In **PL**→**SG**, these

targets reverse. Proximity and majority number require separate incongruent splits because they isolate different attractor patterns. We therefore report each hypothesis with its own pre-intervention counterfactual baseline; the $\mathcal{H}_{syn}$ baseline is averaged across the two heuristic splits. Appendix B specifies every attractor layout and gives complete examples.

5 Experimental Setup

The main text reports Gemma-3-12B (Gemma Team et al., 2025) and Qwen-3-8B (Yang et al., 2025); the appendix reports the same analyses for Llama-3.1-8B (Grattafiori et al., 2024) and OLMo-3-7B (Team Olmo et al., 2026). We compare structural probes, Boundless DAS, and Differential Binary Masking (DBM). The probe tests graph recovery; DAS learns a rotated intervention subspace; DBM learns a sparse mask in the original activation basis. Appendix Tables 4, 6, and 9 give all hyperparameters. Experiments ran on one NVIDIA H200 NVL GPU.

Interventions edit the pre-verb token, where the next-token distribution expresses the agreement decision. Source and base examples require opposite counterfactual verb numbers, and evaluation compares singular and plural forms of the same verb. The dashed baselines show how often the unedited model already selects the target. DAS and DBM use identical pairs and targets, so their contrast isolates the intervention parameterization. We train each layer and hypothesis separately and repeat the primary probe, DAS, and MAR analyses with seeds 42, 96, and 1120. Appendix D gives the complete protocol.

6 Structural Probing

Having formalized the competing causal hypotheses in our controlled testbed, we first ask whether their variables are encoded in the LMs’ representations. Following Hewitt and Manning (2019), we test whether these structures are recoverable without causal intervention.

We first test whether a separate graph for each hypothesis is recoverable without intervention. For sentence x , let $\mathbf{h}_i^{(\ell)}$ be the layer- ℓ hidden state of token i . A linear map B_ℓ predicts squared token distance:

$$\Delta_{B_\ell}(i, j)^2 = \|B_\ell(\mathbf{h}_i^{(\ell)} - \mathbf{h}_j^{(\ell)})\|_2^2$$

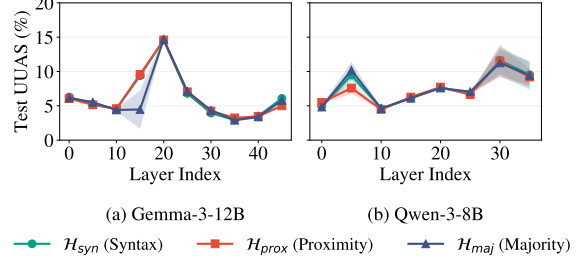


Figure 1: Test UUAS across layers for Gemma-3-12B and Qwen-3-8B. All three hypothesis-specific probes remain low and follow similar trends. Shading shows standard deviation over seeds 42, 96, and 1120; Table 5 reports the values, and Figure 12 shows Llama-3.1-8B.

We fit one probe per hypothesis to its target distance matrix: the dependency tree for $\mathcal{H}_{syn}$, and graphs that attach the target verb to the closest or majority-number attractor for $\mathcal{H}_{prox}$ and $\mathcal{H}_{maj}$. We recover a minimum spanning tree from the predicted distances and report undirected unlabeled attachment score (UUAS). Appendix C gives the graph construction and training details.

Tree Recovery Is Low, but Scalar Features Are Accessible. Figure 1 shows low, non-separating UUAS curves for all three target graphs. Unlike prior probing results on natural corpora (Hewitt and Manning, 2019; Simon et al., 2024, 2025), our controlled setting yields no usable hypothesis-specific tree geometry. This complements Maudslay and Cotterell (2021), who find that probes degrade on Jabberwocky sentences with nonce content words. The comparison suggests that probe success can depend partly on lexical, semantic, or corpus-level regularities that SYNTHAX removes. This is not a failure of the implementation: on the Universal Dependencies English Web Treebank, the same pipeline exceeds 60% peak UUAS in early layers (Figure 13). Nor does low UUAS mean that the three agreement features are absent. Logistic classifiers on Gemma-3-12B’s final-token state predict $\mathcal{H}_{syn}$, $\mathcal{H}_{prox}$, and $\mathcal{H}_{maj}$ number with 97.79%, 98.93%, and 98.95% accuracy, respectively (Figure 14). Together, these checks show a specific gap: the scalar number features are linearly decodable, but the corresponding hypothesis-specific graph geometries are not recovered by this structural probe. Since probes are correlational, this failure does not prove that syntactic variables are absent; it only shows that they are not recoverable by this diagnostic, motivating causal interventions in §7.

7 Causal Interventions: Boundless DAS vs. DBM

Since structural probing evaluates recoverability rather than causal use, we next intervene on the LM’s representations. We ask whether each agreement variable can steer the model, and whether it is accessible in the original activation basis or only after rotation.

Boundless DAS. For base sentence b and source sentence s , let $\mathbf{h}_b^{(\ell)}$ and $\mathbf{h}_s^{(\ell)}$ be their states at layer ℓ and the pre-verb token. Boundless DAS learns an orthogonal rotation R_θ and a scalar boundary that defines a soft mask $\mathbf{m}_\theta$. Fixing the lower boundary at zero makes the mask select a prefix of the rotated space; annealing its temperature turns that soft boundary into a hard m -dimensional selection. The intervention swaps the selected source dimensions into the base state before applying the inverse rotation:

$$\tilde{\mathbf{h}}_b = R_\theta^\top ((1 - \mathbf{m}_\theta) \odot R_\theta \mathbf{h}_b + \mathbf{m}_\theta \odot R_\theta \mathbf{h}_s).$$

The rotation and mask minimize cross-entropy to the counterfactual verb number prescribed by hypothesis $\mathcal{H}$. Interchange Intervention Accuracy (IIA) is the fraction of test pairs for which the intervened model chooses that target. After training, the hard mask selects an m -dimensional subspace whose orthonormal basis is the first m columns of $R_\theta^\top$. Appendix D gives the optimization and pairing details.

Writing $y_{\mathcal{H}}(b, s)$ for the counterfactual target and $\hat{y}_\theta(b, s)$ for the number selected after intervention, the optimization and evaluation are:

$$\mathcal{L}_{\text{DAS}} = \sum_{(b,s)} \text{CE}(p_\theta(\cdot \mid b, s), y_{\mathcal{H}}(b, s)),$$

$$\text{IIA}(\mathcal{H}) = \frac{1}{|S_{\text{test}}|} \sum_{(b,s)} \mathbf{1}[\hat{y}_\theta(b, s) = y_{\mathcal{H}}(b, s)].$$

Thus, IIA measures behavioral control on held-out sentence pairs, not similarity between the learned state and an annotated representation. A high value says that the optimized subspace supports the prescribed counterfactual transformation. Whether that transformation is distinctive must be judged by competing targets, controls, and subspace geometry.

Differential Binary Masking. DBM uses the same source–base pairs and counterfactual targets

but does not learn a rotation. It learns an element-wise gate $\mathbf{g}_\phi = \sigma(\mathbf{m}_\phi/T)$ directly in the original activation basis:

$$\tilde{\mathbf{h}}_b = (1 - \mathbf{g}_\phi) \odot \mathbf{h}_b + \mathbf{g}_\phi \odot \mathbf{h}_s,$$

$$\mathcal{L}_{\text{DBM}} = \text{CE}(p_\phi, y_{\mathcal{H}}) + \lambda \|\mathbf{m}_\phi\|_1.$$

Annealing T turns the gate into a binary mask, while the L_1 term favors a sparse set of original coordinates. Appendix E gives the pair construction, optimization details, hyperparameters, and extended results. The DAS–DBM comparison therefore asks whether the intervention is recoverable only after rotation or through sparse coordinates in the model’s native basis.

Random-Network Control. Because DAS is highly expressive and can fit arbitrary input–output mappings (Arora et al., 2024), we apply the same protocol to a randomly initialized Qwen-3-8B model. We observe that IIA stays near the 50% chance level across layers, directions, and hypotheses (Figure 17), indicating that high IIA in Figure 2 depends on pretrained representational structure rather than DAS alone. This control rules out high IIA arising from the same optimizer on an untrained network. It does not show that a trained-model alignment is unique, necessary during ordinary prediction, or specific to syntax; the competing hypotheses and MAR address those separate questions.

Competing Hypotheses Are Intervention-Compatible. The top panel of Figure 2 shows high-scoring intervention subspaces for all three hypotheses in several settings. $\mathcal{H}_{\text{syn}}$, $\mathcal{H}_{\text{prox}}$, and $\mathcal{H}_{\text{maj}}$ are not minor variants of the same explanation: for the same base sentence, they can prescribe different counterfactual verb-number targets. For Gemma-3-12B, IIA is high across much of the network for both intervention directions and often exceeds the pre-intervention counterfactual baseline by a large margin. Qwen-3-8B is more direction-dependent, with SG→PL favoring $\mathcal{H}_{\text{prox}}$ and $\mathcal{H}_{\text{maj}}$, and PL→SG showing stronger alignment for $\mathcal{H}_{\text{syn}}$. The common result is that even conflicting causal hypotheses can be aligned with intervention subspaces. Therefore, high IIA for $\mathcal{H}_{\text{syn}}$ alone would not establish a unique hierarchical explanation. DAS shows that a hypothesis is intervention-compatible with the model; competing hypotheses must then be evaluated side by side and compared through



Figure 2: A direct comparison of layer-wise IIA in the 4-attractor setting. **Top:** Boundless DAS learns rotated subspaces that substantially exceed the hypothesis-specific pre-intervention baselines in several settings. **Bottom:** DBM applies the same intervention targets in the original activation basis and remains near those baselines. Curves show means over seeds 42, 96, and 1120; shading shows standard deviation. Tables 7 and 8 report the DAS values.

controls and subspace geometry. Llama-3.1-8B and OLMo-3-7B show the same central pattern in Figure 15.

Interpreting IIA Against Its Baseline. The dashed lines in Figure 2 show how often the unedited model already selects each counterfactual target. IIA should be read as a change from that starting point, not as a raw score alone. A high post-intervention value is strong evidence of control when the corresponding baseline is low; the same value is less informative when the model already prefers the target. This comparison is especially important because the two directions reverse the desired verb number and expose different singular–plural preferences. DAS and DBM use the same pairs, targets, and baselines, so their gap cannot be attributed to different evaluation tasks.

The Learned Directions Are Not Sparse Original-Basis Coordinates. The bottom panel of Figure 2 shows the DBM control. DBM rarely rises meaningfully above the pre-intervention baseline for any hypothesis, whereas DAS often reaches high IIA on the same source–base pairs and targets. The comparison therefore isolates the parameterization: DBM can exchange sparse original-basis coordinates, while DAS can first rotate the representation and then exchange a contiguous subspace. The result does not imply that agreement number is ab-

sent from individual activations; the scalar probes in Figure 14 show that all three number labels are decodable. Instead, sparse coordinate exchange is insufficient to reproduce the counterfactual behavior found after rotation. This pattern persists across attractor counts (Figure 18) and for Llama-3.1-8B and OLMo-3-7B (Figures 19 and 20).

Robustness Checks. We vary the attractor count to test whether each learned alignment remains stable as the agreement context deepens. Figure 3 reports peak IIA for one to four attractors. For both models, the relative behavior of each hypothesis is largely preserved as attractor count changes: hypotheses with high IIA at one attractor generally remain high with more attractors. Gemma-3-12B is especially stable across hypotheses and directions. Qwen-3-8B shows more variation in absolute IIA, but its hypothesis-specific profiles are mostly preserved. Thus, the DAS findings are not tied to a single attractor setting. The corresponding Llama-3.1-8B and OLMo-3-7B curves are in Figure 16. A trained-model control that scrambles source–base pairings while preserving the source’s target number also retains high IIA (Figure 4). This transfer across unrelated lexical contexts argues against memorization of sentence pairs, although the control does not rule out every form of optimizer flexibility.

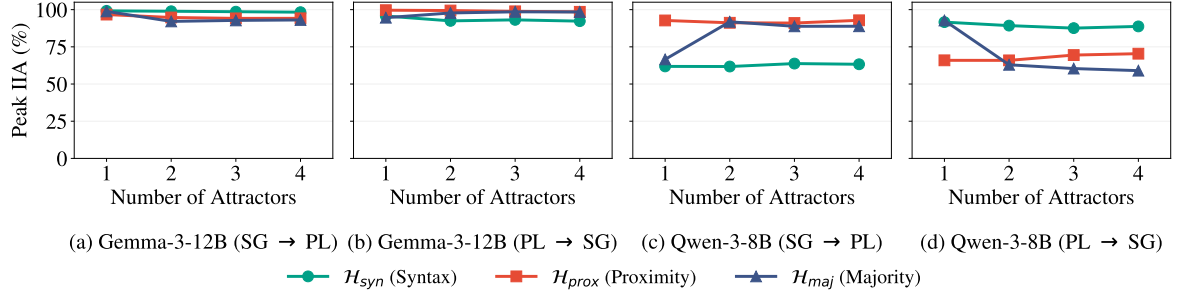


Figure 3: Peak Boundless DAS IIA as the attractor count increases from one to four. Gemma-3-12B remains stable across hypotheses and directions; Qwen-3-8B retains its broad hypothesis profile with greater variation. The result shows that the main intervention pattern is not confined to one dependency depth. Figure 16 gives the corresponding Llama-3.1-8B and OLMo-3-7B results.

The random-network and derangement controls test different failure modes. The random network asks whether DAS can produce high IIA without pretrained parameters; it cannot. Derangement keeps the source’s target number but breaks the original lexical pairing, testing whether the intervention memorizes particular sentence pairs. High IIA after derangement shows that number transfer generalizes across lexical contexts. Because the replacement source still carries the intended number, this control does not determine whether the learned subspace represents the full causal rule or only a transferable number feature.

Subspace Size Depends on Intervention Direction. The dimensionality required by Boundless DAS depends on the target verb’s grammatical number. Forcing a counterfactual plural prediction requires much larger subspaces, exceeding 1,200 dimensions for Gemma-3-12B. Conversely, forcing a singular prediction requires only a fraction of those dimensions. Figure 5 reports both absolute dimension and fraction of hidden size: the Gemma-3-12B syntax subspace for SG→PL spans 32.2% of its hidden size. The asymmetry mirrors the singular preference visible in several pre-intervention baselines (Li et al., 2023): plural-target interventions often have farther to move the output distribution. Subspace dimension is still an outcome of the DAS optimization, not a direct measure of a rule’s complexity or causal importance.

8 Quantifying Subspace Overlap

High IIA establishes that a hypothesis can be implemented by an intervention, but not whether its directions are distinct from those of a competing hypothesis. We therefore compare the learned DAS subspaces directly.

Manifold Alignment Ratio. Let $W_{syn} \in \mathbb{R}^{d \times m_{syn}}$ and $W_{heur} \in \mathbb{R}^{d \times m_{heur}}$ contain orthonormal bases for the syntax and heuristic subspaces. We define:

$$\text{MAR}(W_{syn}, W_{heur}) = \frac{\|W_{syn}^\top W_{heur}\|_F^2}{\sqrt{m_{syn}m_{heur}}}. \quad (1)$$

Appendix F.1 shows that the numerator is the sum of squared cosines of the principal angles, so the score depends on the subspaces rather than a particular choice or ordering of basis vectors. Higher MAR means that two interventions reuse more directions; lower MAR means greater separation. Zero denotes orthogonality. Because attainable overlap depends on subspace size, we interpret each score against a dimension-matched random baseline.

Model and Direction Matter. Table 1 shows that every empirical overlap exceeds its matched null, so subspace dimension alone does not explain the shared directions. Gemma has greater syntax–heuristic overlap in three of the four comparisons, whereas Qwen’s largest overlap is with majority number for SG→PL; for this heuristic, Qwen rises from 0.3373 for PL→SG to 0.4281 for SG→PL, while Gemma falls from 0.5570 to 0.3393. The substantial standard deviations, especially for Qwen, support broad patterns rather than precise condition rankings. MAR therefore depends on the model, rule, and intervention direction; Appendix F.2 reports the Llama-3.1-8B and OLMo-3-7B results.

What MAR Adds to IIA. IIA asks whether an intervention reaches the output required by one rule. MAR asks whether two successful interventions use the same rotated directions. Two hypotheses can therefore achieve similar IIA even when

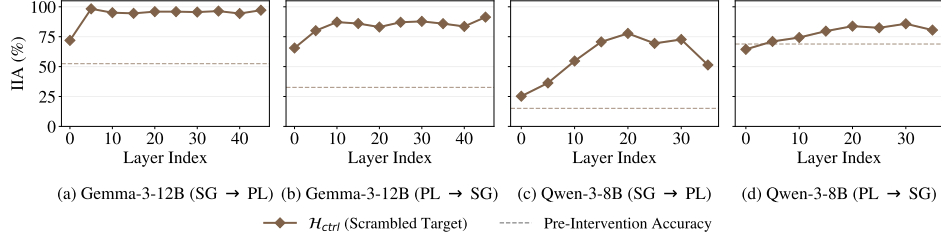


Figure 4: IIA after deranging source–base pairings within each target-number split. Successful transfer across unrelated lexical contexts argues against memorization of particular sentence pairs. Because the replacement source preserves the intended target number, this is a partial control for lexical correspondence rather than a proof that the learned alignment is unique.

Model	Direction	MAR_{prox} (Null)	MAR_{maj} (Null)
Gemma-3-12B	PL → SG	0.4608 ± 0.1596 (0.0702)	0.5570 ± 0.1968 (0.1419)
	SG → PL	0.3907 ± 0.0783 (0.1211)	0.3393 ± 0.0685 (0.1186)
Qwen-3-8B	PL → SG	0.3574 ± 0.2221 (0.0895)	0.3373 ± 0.2253 (0.1113)
	SG → PL	0.2571 ± 0.0819 (0.1089)	0.4281 ± 0.2505 (0.1056)

Table 1: Mean $\pm$ standard deviation of MAR over three seeds at the selected 4-attractor layers. Parentheses give the mean of the dimension-matched random-subspace null. All empirical values exceed their nulls with right-tailed $p < 10^{-10}$; Appendix Table 12 gives the full tests, and Table 11 gives all four models.

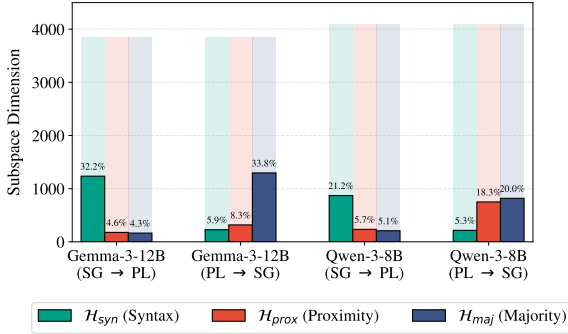


Figure 5: Learned subspace dimensions at the best 4-attractor intervention layers. Percentages give the fraction of hidden size (3,840 for Gemma-3-12B; 4,096 for Qwen-3-8B). Steering toward plural generally uses more dimensions than steering toward singular. Figure 21 gives Llama-3.1-8B and OLMo-3-7B.

their interventions act through different subspaces. Conversely, above-null MAR shows geometric reuse even when the hypotheses prescribe different outputs for the same sentence. High IIA with low MAR would reveal distinct ways to steer the output; high IIA with above-null MAR, as observed here, shows that conflicting rules reuse part of the same intervention geometry. This does not mean that the unedited model follows both rules. It means that intervention success alone cannot distinguish them.

Shared Directions Do Not Imply Shared Rules. The hypotheses disagree about which noun deter-

mines the verb, but every intervention changes the same binary choice between singular and plural forms. Their overlap may therefore reflect shared number-control directions, with rule-specific information elsewhere in each subspace. However, variation across heuristics and directions suggests that model- and rule-specific factors also contribute; the experiments do not isolate these sources. The results establish only that successful syntax and heuristic interventions overlap more than matched random subspaces. Combined with high IIA for the competing rules, this makes syntax causally implementable but not uniquely identified.

9 Conclusion

Our results show a gap between recoverability and intervention compatibility. Structural probes fail to recover hypothesis-specific structure, but Boundless DAS learns subspaces that steer predictions according to $\mathcal{H}_{syn}$, $\mathcal{H}_{prox}$, and $\mathcal{H}_{maj}$. DBM stays near baseline, so these directions are accessible after rotation but not as sparse original-basis coordinates. MAR shows that syntax and heuristic interventions overlap more than matched random subspaces. High IIA shows a subspace supports an intervention, not that the hypothesis uniquely explains the model; MAR complements IIA by testing whether alternatives occupy distinct or overlapping subspaces.

Limitations

Our evidence is restricted to synthetic English subject–verb agreement. Repeated relative-clause templates can create template- or position-specific cues, even with disjoint test vocabulary. We do not evaluate a held-out-template split, a semi-natural agreement set, other syntactic phenomena, or other languages. The natural-text experiment validates the structural-probing pipeline, not the paper’s causal conclusions. The results should therefore be read as a controlled case study of ambiguity among three agreement rules, not as a general account of hierarchical syntax in LMs.

The hypothesis set is not exhaustive: models may combine the tested rules or use lexicalized, positional, attention-based, or nonlinear cues. All interventions target the pre-verb token, and no position ablation is available. Boundless DAS searches rotated linear subspaces; this constraint avoids unrestricted nonlinear alignments that can make causal abstraction uninformative (Sutter et al., 2026), but it can miss nonlinear or multi-position variables. DBM tests sparse masks only in the original basis, so its failure does not prove that no localized mechanism exists under another parameterization.

MAR measures overlap between learned intervention subspaces, not causal necessity, model quality, or out-of-distribution generalization. Although we report random-subspace nulls and three optimization seeds, a same-hypothesis retraining baseline would better calibrate how much variation arises even when the target explanation is unchanged. Finally, the four evaluated models range from 7B to 12B parameters; different scales or architectures may yield different causal geometries.

Ethical Considerations

SYNTAX contains synthetic sentences about formal agreement and no personal data, which limits direct privacy and toxicity risks. The intervention methods could nevertheless be repurposed to steer harmful behavior, and the layer-wise experiments consume substantial accelerator resources. An interpretability claim based on an insufficiently tested causal mechanism could also mislead downstream safety evaluations. Testing competing explanations and reporting non-unique alignments may reduce this risk by making mechanistic claims easier to challenge and falsify.

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• **Appendix E: Differential Binary Masking (DBM).** *Defines the original-basis control and reports extended model and attractor results.*

• **Appendix F: Manifold Alignment Ratio (MAR).** *Proves the principal-angle formulation and reports dimension-matched null and seed analyses.*

A Formalization of Causal Hypotheses

We formalize the three hypotheses as causal models. A causal model $\mathcal{M}$ is a pair $(\mathcal{S}, \mathcal{F})$, where $\mathcal{S}$ is a signature and $\mathcal{F}$ associates a structural equation $\mathcal{F}^X$ with every variable X . The signature $\mathcal{S}$ consists of variables $\mathcal{V}$ and a function $\mathcal{R}$ that gives the possible values of each variable. An intervention $X \leftarrow x$ replaces $\mathcal{F}^X$ with the constant value x .

For our Subject-Verb Agreement task, we define a shared signature for all three hypotheses. Let $\mathcal{V}_{input} = \{N_{subj}, N_{attr_1}, \dots, N_{attr_k}\}$ represent the morphological number feature of the nouns in the context, where $\mathcal{R}(N) \in \{0, 1\}$ (mapping to Singular and Plural, respectively). Let O represent the final causal output variable determining the verb’s number, where $\mathcal{R}(O) \in \{0, 1\}$.

The structural equations $\mathcal{F}^O$ strictly differentiate the three causal models:

1. Hierarchical Syntax ($\mathcal{H}_{syn}$): The causal output is strictly determined by the structural root, independent of any intermediate variables in the linear sequence.

$$\mathcal{F}_{syn}^O = N_{subj}$$

2. Proximity Recency ($\mathcal{H}_{prox}$): The causal output is determined exclusively by the final input variable in the linear sequence, representing the immediately adjacent noun.

$$\mathcal{F}_{prox}^O = N_{attr_k}$$

3. Majority Number ($\mathcal{H}_{maj}$): The causal output is the majority number among the intervening attractor nouns; the structural main subject is excluded from this surface heuristic.

$$\mathcal{F}_{maj}^O = \mathbb{I} \left(\sum_{i=1}^k N_{attr_i} > \frac{k}{2} \right)$$

where $\mathbb{I}$ is the indicator function. By performing interchange interventions on the variables defined in these structural equations, we can systematically verify whether the targeted neural representations mirror the causal properties of $\mathcal{H}_{syn}$, $\mathcal{H}_{prox}$, or $\mathcal{H}_{maj}$.

B SYNTHAX: Dataset Construction and Lexical Curation

B.1 Structural Templates and Semantic Control

SYNTHAX restricts generation to object-relative subject–verb agreement templates adapted from [Marvin and Linzen \(2018\)](#). This controlled scope reduces, but cannot eliminate, semantic and distributional alternatives to the three tested agreement rules.

We established two distinct structural templates to maintain domain diversity without introducing selectional restriction violations:

• **Animate template:**

[D, MS, C, D, ES, EV, MV]

• **Inanimate template:**

[D, IS, IC, D, ES, EV, IV]

Table 2 details the specific syntactic components comprising these templates. To maximize simplicity and ensure strict semantic coherence across all recursive depths, we deliberately restricted the vocabulary of specific functional tokens. Determiners (D) are exclusively limited to the definite article (“the”). Furthermore, to avoid grammatically awkward generations, complementizers are strictly mapped to the animacy of their preceding noun: animate subjects (C) may take *who*, *whom*, or *that*, whereas inanimate subjects (IC) are restricted exclusively to *that*.

Figure 6 shows how the templates recursively stack embedded clauses. Separating animate and inanimate lexical items reduces obvious selectional violations; it does not guarantee that models ignore every semantic cue.

B.2 Diagnostic Configurations and Attractor Layouts

The data splits create controlled conflicts among the three target rules before any intervention is applied.

Below, we give the sequence layouts and 3-attractor examples. **Main** denotes the main subject and **Attr.** an intervening attractor.

1. Proximity Recency ($\mathcal{H}_{prox}$):

This setup tests whether the model simply copies the morphological number of the noun linearly adjacent to the main verb.

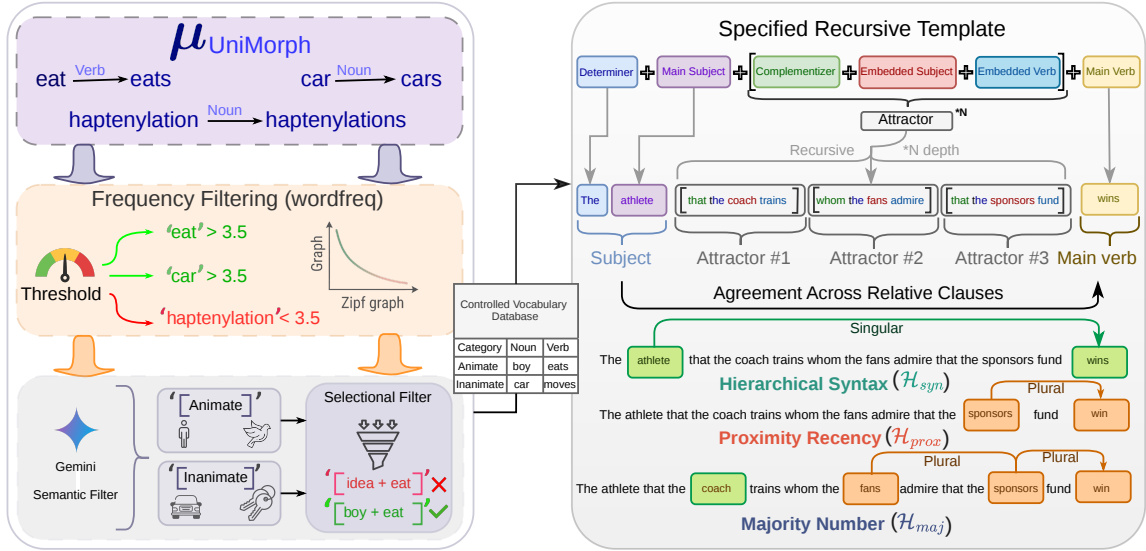


Figure 6: The SYNTHAX pipeline. Lexical filters control morphology, frequency, and semantic compatibility; recursive relative-clause templates then vary attractor number so that structural, proximity, and majority-number rules can be assigned conflicting targets.

Component	Sym.	Description	Vocab / Example
Determiner	D	Definite article	<i>the</i>
Main Subject	MS/IS	Head noun (Anim/Inanim)	<i>athlete / car</i>
Complementizer	C/IC	Relative pronoun	<i>who, whom, that / that</i>
Embedded Subject	ES	Intervening attractor	<i>coach / machine</i>
Embedded Verb	EV	Verb of relative clause	<i>trains / operates</i>
Main Verb	MV/IV	Target verb	<i>wins / breaks</i>

Table 2: Syntactic components and vocabulary restrictions for the SYNTHAX generation templates.

SG Setup:

Main ... Attr. ... Attr. → Target Verb
SG N-1 (SG) 1 (PL)

Example (3 Attr.): The athlete that the coach trains whom the manager directs that the **sponsors** fund **wins/win**.
(Syntax targets “wins” [SG]; Proximity targets “win” [PL])

PL Setup:

Main ... Attr. ... Attr. → Target Verb
PL N-1 (PL) 1 (SG)

Example (3 Attr.): The athletes that the coaches train whom the managers direct that the **sponsor** funds **win/wins**.
(Syntax targets “win” [PL]; Proximity targets “wins” [SG])

2. Majority Number ($\mathcal{H}_{maj}$):

This setup tests whether the model follows the

majority number among the intervening attractor nouns, excluding the structural main subject.

SG Setup:

Main ... Attr. ... Attr. → Target Verb
SG N-1 (PL) 1 (SG)

Example (3 Attr.): The athlete that the **fans** admire whom the **sponsors** fund that the coach trains **wins/win**.
(Syntax targets “wins” [SG]; Majority targets “win” [PL])

PL Setup:

Main ... Attr. ... Attr. → Target Verb
PL N-1 (SG) 1 (PL)

Example (3 Attr.): The athletes that the **fan** admires whom the **sponsor** funds that the coaches train **win/wins**.
(Syntax targets “win” [PL]; Majority targets “wins” [SG])

3. Hierarchical Syntax ($\mathcal{H}_{syn}$):

This hypothesis posits that the language model processes text using a structural dependency tree. Under this baseline, the morphological number of the target verb must always agree with the structural root (the Main Subject), completely ignoring any intermediate attractors regardless of their linear position or frequency.

For both the $\mathcal{H}_{prox}$ and $\mathcal{H}_{maj}$ splits, the $\mathcal{H}_{syn}$ target changes only the main verb so that it agrees with the main subject; all attractors remain unchanged. We macro-average structural accuracy across the proximity and majority splits.

B.3 Dataset Scaling, Diversity, and Tokenization Constraints

We constrain the lexicon to reduce direct reuse of vocabulary and local word sequences.

- **Data Splits and Balancing:** We generated a total of 20,000 training instances, perfectly balanced between 10,000 animate and 10,000 inanimate sentences. We reserved 1,000 instances strictly for the validation set.
- **Evaluation Set:** We generated a separate, held-out test set of 3,000 instances (1,500 animate, 1,500 inanimate). *It is critical to note that all diagnostic plots, metric evaluations, and subspace visualizations presented in this work are computed exclusively on this test set.*
- **Lexical Rarity and Strict Disjointness:** Across all dataset partitions, we restrict nouns and verbs to a Zipf frequency score ≥ 3.5 (Figure 6). Test nouns and verbs are disjoint from those in training and validation, which blocks direct lexical reuse across the split.
- **Test Lexicon:** Gemini (Comanici et al., 2025) filters the held-out vocabulary for less predictable nouns and verbs. This reduces common collocations but does not force a model to use hierarchy.
- **Intra-sentence Diversity:** We generated diverse combinations of noun and verb pairings across training, validation, and testing partitions for all attractor counts ($N = 1$ to 4). Furthermore, we enforced a strict non-repetition constraint: within any single sentence, no two nouns or verbs are identical. The main subject, attractors, and all verbs are guaranteed to be unique lexical items.

- **Tokenization Compatibility:** We retain only target verbs that form one token under every evaluated model’s tokenizer, avoiding sub-token differences at the intervention target.

B.4 Prompt Templates for Lexical Filtering

Gemini supplies the semantic labels used to filter the vocabulary. The prompts for each stage follow.

Noun Semantic Categorization: The first stage of our pipeline required strictly separating nouns into animate and inanimate categories to prevent semantic anomalies during sentence generation. Figure 8 displays the prompt used to classify the base noun vocabulary, ensuring that subsequent structural templates align with logical real-world properties.

Verb Coherence Filtering: To eliminate selectional restriction violations (e.g., “the idea eats”), verbs had to be explicitly matched to appropriate noun categories. Figure 9 illustrates the prompt used to filter verbs based on their semantic compatibility with either animate subjects, inanimate subjects, or both.

Adversarial Noun Curation (Test Set): The held-out test vocabulary is disjoint from training and validation. Figure 10 gives the prompt used to select its less predictable nouns.

Adversarial Verb Curation (Test Set): Figure 11 gives the test-verb prompt, which excludes common and highly irregular forms (such as forms of *be*, *have*, and *do*) and ambiguous parts of speech.

B.5 Sample Sentences from the SYNTHAX Dataset

Table 3 presents 16 generated examples from the $N = 3$ attractor setting, split across animate and inanimate templates. To illustrate how SYNTHAX separates competing causal mechanisms, we group examples by target hypothesis and highlight the lexical items that determine the required verb number: the structural root for $\mathcal{H}_{syn}$, the adjacent noun for $\mathcal{H}_{prox}$, and the majority-number nouns for $\mathcal{H}_{maj}$.

B.6 Validation of Gemini-Based Semantic Filtering

We evaluate the automatic lexical labels with an expert-annotated sample and an automated WordNet comparison. These checks estimate filtering accuracy; they cannot rule out all semantic confounds.

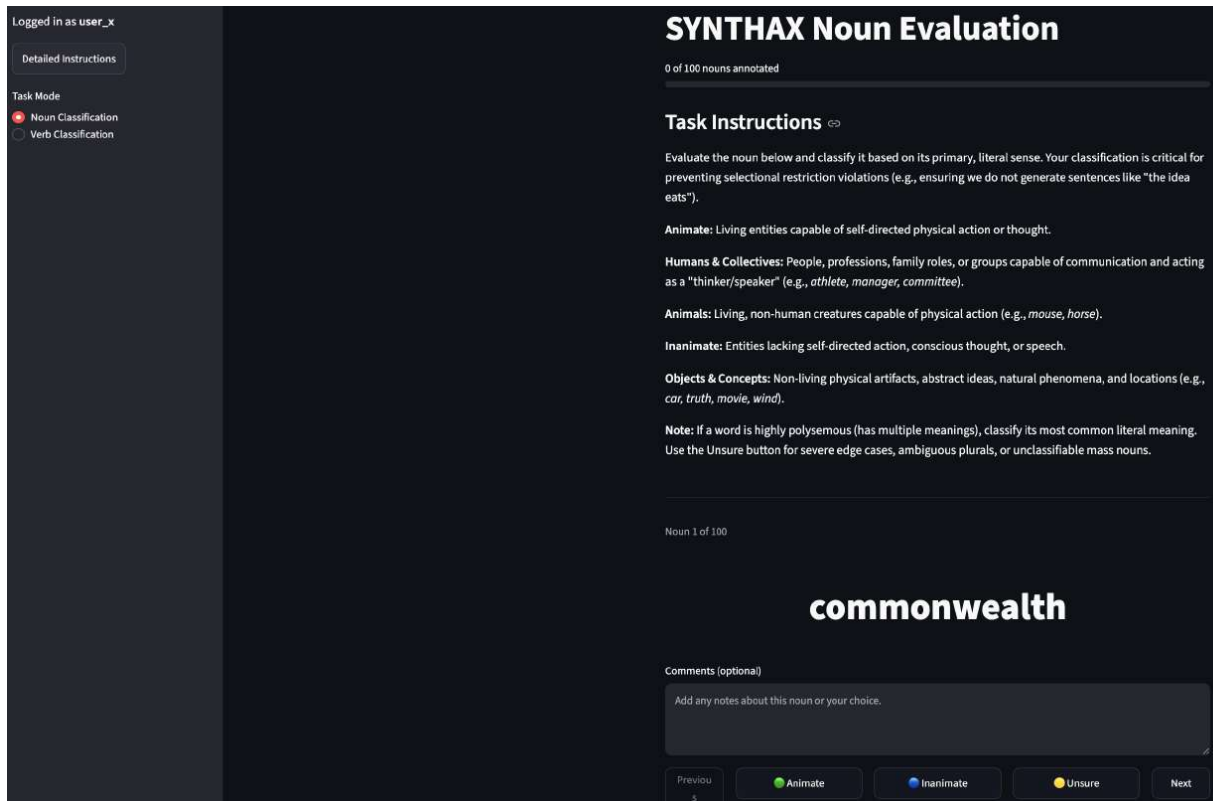


Figure 7: Screenshot of the User Application developed for validating of Gemini-Based Semantic Filtering

Human Evaluation and Inter-Annotator Agreement (IAA)

To validate the model’s semantic classification against a gold-standard linguistic baseline, we conducted a blind human evaluation of the filtered vocabulary.

- **Sampling and Annotation:** We uniformly sampled 100 nouns and 100 verbs from the generated test-set vocabulary. Five annotators with PhDs in Linguistics independently evaluated these items using concise on-screen guidelines and a detailed reference document.
- **Reference Labels:** The majority vote of the five annotators defines the reference label for each item.
- **Model Accuracy:** Against this consensus, Gemini reaches 96.0% accuracy for noun animacy and 98.0% for verb categorization.
- **Annotator Agreement:** Fleiss’ κ is 0.88 for noun animacy and 0.92 for verb assignments.

Automated WordNet Verification

We also compare the filtered test vocabulary with WordNet.²

- **Noun Animacy Alignment:** We mapped the primary synset categories (lexname) of our sampled nouns to verify strict animacy constraints. Nouns classified by the model as animate were cross-checked against WordNet categories such as `noun.person` or `noun.animal`, while inanimate nouns were checked against categories like `noun.artifact` or `noun.object`. This automated mapping achieved a 98.0% alignment with WordNet’s established ontology.
- **Verb Transitivity Alignment:** We compare the model’s broad transitivity labels with WordNet syntactic frames. The labels agree on 95.0% of the checked items.

C Structural Probing: Methodology, Training, and Extended Results

C.1 Data Preprocessing and Alignment

We parse each sentence with Stanza (Qi et al., 2020) and convert the output to CoNLL-U. The depen-

²<https://github.com/globalwordnet/english-wordnet>

Prompt for Noun Filtering (animate/inanimate)

You are an expert Computational Linguist creating a high-quality dataset for a Subject-Verb Agreement experiment. Your goal is to filter a raw, noisy list of words and classify them into strict semantic categories with specific template tags.

Your Task: Process the raw list of words below. For each word:

Clean it: Extract the strict Lemma (Singular) and Form (Plural). If only one is provided, generate the missing form.

Classify it: Assign it to strictly one of two categories (ANIMATE or INANIMATE).

Tag it: Assign the specific template tags (MS, ES, BS, IS) based on the logic below.

1. Classification Rules (The Logic)

You must classify every valid noun into exactly one of these two categories:

Category A (a): ANIMATE (Human / Profession)

Definition: Humans, professions, family roles, or distinct groups of people capable of communication and complex thought.

Template Tags: ["MS", "ES", "BS"]

Reasoning: Humans can be the Main Subject (MS), Embedded Subject (ES), and the "Thinker/Speaker" (Base Subject BS) for sentential complements (e.g., "The senator claimed that...").

Category A (b): ANIMATE (Animal)

Definition: Living animals that are not humans.

Template Tags: ["MS", "ES"]

Reasoning: Animals can be subjects, but generally cannot function as the "Base Subject" (BS) for verbs like claimed, asserted, wrote, said.

Category B: INANIMATE NOUNS

Definition: Non-living physical objects, artifacts, locations, abstract concepts, or natural phenomena.

Template Tags: Assign ["IS"].

Reasoning: Inanimate nouns serve as the Inanimate Subject (IS). They generally cannot be Base Subjects because they cannot "think" or "say" things.

Examples: mile, road, idea, movie, rifle, table, wind.

2. Filtering Rules (Strictly Discard These)

If a word belongs to any of the following categories, SKIP IT ENTIRELY. Do not include it in the output.

Wrong POS: Verbs (e.g., eat, tempting), Adjectives (chaotic, focal), or Adverbs.

Proper Nouns: Specific names of people or places (e.g., Daniel, London, Eden's, Soviet). We only want Common Nouns.

Slang / Offensive: Informal, rude, or highly colloquial terms (e.g., psycho, chick, motherfucker). Keep the tone formal and standard.

Ambiguous Plurals: Words where the Singular and Plural forms are identical (e.g., sheep, fish, moose). These ruin agreement experiments.

3. formatting Requirements

Input Handling: The input list is messy (e.g., mile (miles) N; PL). You must extract the clean Lemma (Singular) and Form (Plural).

Auto-Inflection: If the input only gives the singular, you must generate the correct plural form.

Output Format: strict JSONL. One valid JSON object per line. No markdown code blocks, no intro text.

Example Output (Your Target):

```

JSON
{"lemma": "mile", "plural": "miles", "category": "INANIMATE", "tags": ["IS"]}
{"lemma": "manager", "plural": "managers", "category": "ANIMATE", "tags": ["MS", "ES", "BS"]}
{"lemma": "mouse", "plural": "mice", "category": "ANIMATE", "tags": ["MS", "ES"]}
{"lemma": "cat", "plural": "cats", "category": "ANIMATE", "tags": ["MS", "ES"]}
Raw Word List to Process: [PASTE YOUR BATCH OF WORDS HERE]

```

Figure 8: Prompt used to classify training and validation nouns as animate or inanimate.

Prompt for Verb Filtering (animate/inanimate)

You are an expert Computational Linguist. Your task is to classify a raw list of verbs into strict syntactic categories (Slots) based on the **Recursive Grammar Templates** provided below.

****Context: The Template Architecture****

We are building a dataset using the following Python template rules. A verb's "Tag" is determined by its position and function in these specific structures.

****Your Task:****

Process the raw verb list below. For each verb, extract the **Singular (3rd Person)** and **Plural (Base)** forms, and assign strict tags based on the logic above.

**The Classification Rules (Test each verb against these Templates)**

****1. Tag: `MV` (Intransitive Human Action)****

* **Template Test:** `simple\_agrmt`: ['D', 'MS', 'MV']

* **Context:** "The **author** [VERBS]." (Must be complete without an object).

* **Examples:** laughs, smiles, runs, sleeps, waits, speaks.

* **Discard if:** It needs an object (e.g., likes, puts).

****2. Tag: `IV` (Intransitive Object Action)****

* **Template Test:** `obj\_rel\_across\_inanim`: ['D', 'IS', ..., 'IV']

* **Context:** "The **movie** [VERBS]." (Logic: Guards [VERB] Author).

* **Examples:** ends, starts, falls, burns, breaks, shines.

****3. Tag: `EV\_ANIM` (Transitive Action on Human)****

* **Template Test:** `obj\_rel\_across\_anim`: [..., 'MS', ..., 'EV']

* **Context:** "The **author** that the guards [VERB]..." (Logic: Guards [VERB] Author).

* **Rule:** Must be Transitive. Target is **Human**.

* **Examples:** likes, admires, hates, loves, hugs, criticizes, visits.

****4. Tag: `EV\_INAN` (Transitive Action on Object)****

* **Template Test:** `obj\_rel\_across\_inanim`: [..., 'IS', ..., 'EV']

* **Context:** "The **movie** that the guards [VERB]..." (Logic: Guards [VERB] Movie).

* **Rule:** Must be Transitive. Target is **Inanimate**.

* **Examples:** watched, bought, read, wrote, broke, found.

****5. Tag: `BV` (Mental/Speech Verb)****

* **Template Test:** `sent\_comp`: ['D', 'BS', 'BV', ...]

* **Context:** "The **banker** [VERBS] that..."

* **Examples:** thought, knew, said, claimed, believed, argued, asserted.

****6. Tag: `RMV` (Reflexive Action)****

* **Template Test:** `simple\_reflexives`: [..., 'RMV', 'ANPHR']

* **Context:** "The **author** [VERBS] himself."

* **Examples:** hurt, injured, embarrassed, disguised, cut.

****Output Format:****

Return valid **JSONL**.

* **sg:** 3rd Person Singular (e.g., "eats")

* **pl:** Base Form (e.g., "eat")

* **tags:** List of matching tags ['MV', 'IV', 'EV_ANIM', 'EV_INAN', 'BV', 'RMV']

****Example Output:****

```

```json
{"sg": "laughs", "pl": "laugh", "tags": ["MV"]}
{"sg": "likes", "pl": "like", "tags": ["EV_ANIM", "EV_INAN"]}
{"sg": "reads", "pl": "read", "tags": ["EV_INAN"]}
{"sg": "thinks", "pl": "think", "tags": ["MV", "BV"]}
{"sg": "hurts", "pl": "hurt", "tags": ["RMV", "EV_ANIM", "EV_INAN"]}
```

```

Raw Verb List to Process:\n[PASTE YOUR LIST HERE]

Figure 9: Prompt used to match verbs with compatible animate or inanimate subjects.

Prompt for Noun Filtering for Test Set

You are an expert Psycholinguist and Computational Linguist creating a strict Out-Of-Distribution (OOD) held-out test dataset for a Subject-Verb Agreement experiment.

Your Goal: I will provide you with a raw, noisy batch of 50 nouns. You must act as a strict gatekeeper. You will DISCARD all common, easy, or invalid words, and KEEP ONLY the rare, highly academic, or morphologically irregular nouns.

Your Task: Process the raw list below. Evaluate each word against the strict selection criteria. If a word passes the filter, output it in the EXACT SAME JSONL format as requested, without adding any new fields.

1. Selection Rules (KEEP These)

To be selected, a noun MUST fall into one of these two categories:

* Category A: IRREGULAR PLURALS

* Definition: Nouns that do not form their plural by simply adding "-s" or "-es".

* Examples: criterion/criteria, index/indices, ox/oxen, mouse/mice, thesis/theses.

* Category B: RARE / ACADEMIC / COMPLEX

* Definition: Low-frequency, highly domain-specific, or multi-syllabic academic nouns.

2. Rejection Rules (Strictly DISCARD These)

If a word fits any of these descriptions, SKIP IT ENTIRELY. Do not include it in the output.

* Common / High-Frequency: Everyday words (e.g., dog, car, teacher, student, manager). They belong in the training set, not the OOD test set.

* Ambiguous Plurals: Words where Singular and Plural are identical (e.g., sheep, aircraft). These ruin agreement experiments.

* Mass / Uncountable Nouns: Words that do not naturally pluralize (e.g., furniture, dust).

* Wrong POS / Invalid: Verbs, adjectives, proper nouns (names/places), or slang.

3. Output Formatting (CRITICAL)

* Output strictly in JSONL format. One valid JSON object per line.

* Do NOT add any new fields (no 'complexity' or 'reason' fields).

* Output EXACTLY these four keys: "lemma", "plural", "category", "tags".

* No markdown code blocks, no intro text, no conversational filler.

Example Output (Your Target):

```
{"lemma": "criterion", "plural": "criteria", "category": "INANIMATE", "tags": ["IS"]}
```

```
{"lemma": "oligarch", "plural": "oligarchs", "category": "ANIMATE", "tags": ["MS", "ES", "BS"]}
```

```
{"lemma": "matrix", "plural": "matrices", "category": "INANIMATE", "tags": ["IS"]}
```

Raw Word List to Process:

[PASTE YOUR BATCH OF 50 WORDS HERE]

Figure 10: Prompt used to curate the held-out, lexically disjoint test nouns.

Prompt for Verb Filtering for Test Set

You are an expert Psycholinguist and Computational Linguist creating a strict Out-Of-Distribution (OOD) held-out test dataset for a Subject-Verb Agreement experiment.

Your Goal: I will provide you with a raw batch of 50 JSON objects containing English verbs. You must act as a strict gatekeeper. You will DISCARD all common, high-frequency verbs, and KEEP ONLY the rare, highly academic, or structurally ambiguous verbs.

Your Task: Evaluate each verb against the strict selection criteria below. If a verb passes the filter, output it in the EXACT SAME JSONL format as it was provided.

1. Selection Rules (KEEP These)

To be selected, a verb MUST fall into:

* Category A: RARE / ACADEMIC / COMPLEX

* Definition: Low-frequency, highly domain-specific, or multi-syllabic verbs.

2. Rejection Rules (Strictly DISCARD These)

If a verb fits any of these descriptions, SKIP IT ENTIRELY. Do not include it in the output.

* Common / High-Frequency: Everyday, simple verbs (e.g., eats, leads, brings, asks, serves, handles). These belong in the training set, not the OOD test set.

* Highly Irregular "Be/Have/Do": Do not include forms of be, have, or do.

* Invalid POS: If the word cannot logically or grammatically function as a verb in standard English.

3. Output Formatting (CRITICAL)

* Output strictly in JSONL format. One valid JSON object per line.

* Output EXACTLY the same keys as the input: "sg", "pl", "tags".

* Do NOT add any new fields (no 'complexity', 'reason', or 'category' fields).

* No markdown code blocks, no intro text, no conversational filler. Just the raw JSON lines.

Example Input:

{ "sg": "eats", "pl": "eat", "tags": ["MV", "EV_ANIM", "EV_INAN"] }

{ "sg": "wardens", "pl": "warden", "tags": ["EV_ANIM", "EV_INAN"] }

{ "sg": "asks", "pl": "ask", "tags": ["MV", "EV_ANIM", "EV_INAN", "BV"] }

Example Output (Your Target - dropping 'eats' and 'asks', keeping 'wardens'):

{ "sg": "wardens", "pl": "warden", "tags": ["EV_ANIM", "EV_INAN"] }

Raw Word List to Process:

[PASTE YOUR LIST HERE]

Figure 11: Prompt used to filter held-out test verbs by frequency, regularity, and part of speech.

dependency trees supply the token indices, heads, and pairwise path distances used as probe targets. We use the same split as the intervention experiments: 19,000 training, 1,000 validation, and 3,000 test instances.

C.2 Probe Application Across Competing Hypotheses

While Section (§6) outlines the core probing objective (minimizing the difference between predicted squared distances and hypothesis-specific target distances), the practical application required dynamically altering the CoNLL-U dependency structures to construct the target distance matrices ($\Delta_{\mathcal{H}}$) for each specific causal hypothesis:

1. Hierarchical Syntax ($\mathcal{H}_{syn}$): We use the Stanza dependency tree without modification.

2. Proximity Recency ($\mathcal{H}_{prox}$): We manipulated the target distance matrix by severing the verb’s dependency on the main subject. Instead, we artificially reassigned the structural head of the target verb to the linearly closest attractor noun. The graph distances for all other tokens remained unchanged.

3. Majority Number ($\mathcal{H}_{maj}$): We similarly reconstructed the dependency graph to force the target verb to depend on the noun representing the majority morphological number within the context window.

By mapping these distinct graphs back to token-wise distance matrices, we ensured the probe was trained to recover the exact variable prescribed by each causal model.

Furthermore, while our causal intervention experiments (DAS) evaluate both $SG \rightarrow PL$ and $PL \rightarrow SG$ directions, structural probing does not involve counterfactual interventions. Therefore, to evaluate the effect of syntactic distance on structural recoverability, we restricted our base configurations to sentences where the main subject is Singular (SG) and the number of embedded attractors is four.

C.3 Hyperparameters and Training Details

All Probes were trained to minimize the L_1 loss of the predicted squared distance with respect to the true target distance. The maximum rank of the linear transformation matrix B_l was constrained by the native hidden dimension of the respective evaluated language model. For distance probes, the batch loss was normalized by the square of the sentence length ($|x|^2$) and summed across all

sentences prior to gradient updates. At the end of each epoch, the validation (dev) loss was computed; if a new minimum was not achieved, the optimizer was reset (clearing all momentum states) and the learning rate was decayed by a factor of 0.1.

C.4 Probing Results on Llama-3.1-8B

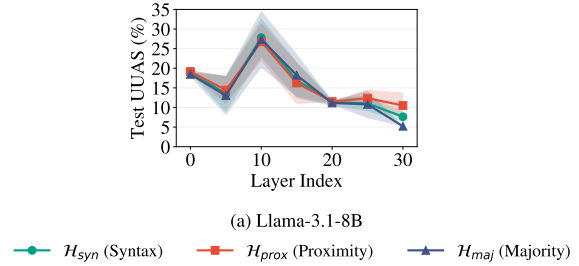


Figure 12: Structural probing results on Llama-3.1-8B across layers. The shaded regions denote the standard deviation across three random seeds (42, 96, 1120); Appendix C.7 reports the exact $\mu \pm \sigma$ values.

Figure 12 shows low UUAS throughout Llama-3.1-8B in the 4-attractor setting, and the curves for $\mathcal{H}_{syn}$, $\mathcal{H}_{prox}$, and $\mathcal{H}_{maj}$ overlap. This matches the Gemma-3-12B and Qwen-3-8B results in §6: this probe does not recover the three target graph geometries in SYNTHAX.

C.5 Sanity Check: Structural Probing on Natural Text

We apply the same distance-probe objective and UUAS evaluation to the standard Universal Dependencies English Web Treebank splits.³ Figure 13 shows peak UUAS above 60% in early layers for Gemma-3-12B, Qwen-3-8B, and Llama-3.1-8B. The pipeline can therefore recover dependency structure in a standard natural-text setting. This check narrows the interpretation of the low SYNTHAX UUAS: it is specific to the controlled data and hypothesis graphs, rather than a general inability of the implementation to reproduce structural probing. It does not show which property of SYNTHAX causes the difference.

C.6 Linear Probing of Scalar Number Features

We also train logistic-regression classifiers on Gemma-3-12B’s final-token representations at layer 45 in the 4-attractor split. Each classifier

³https://universaldependencies.org/treebanks/en_ewt/index.html

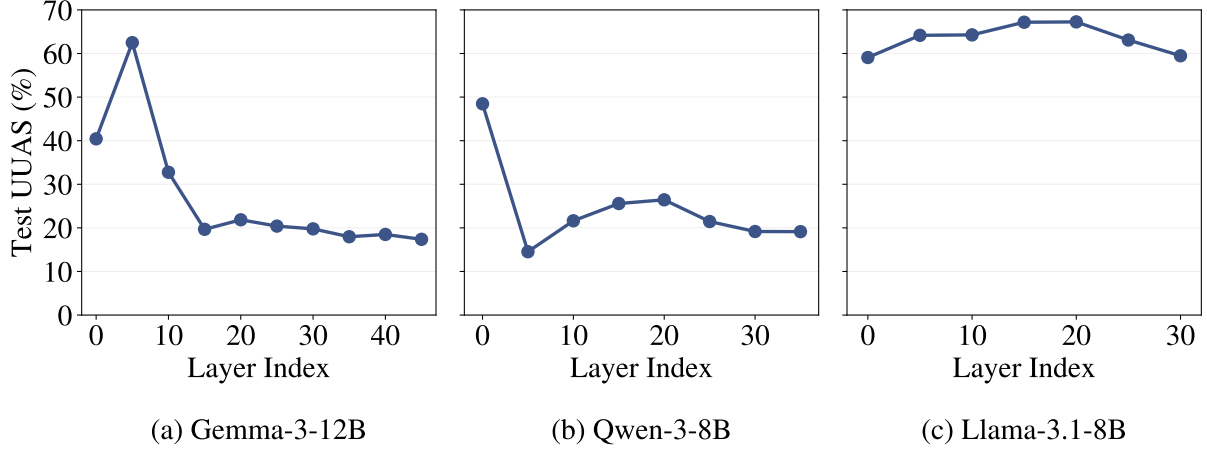


Figure 13: Sanity check results: Structural probing Test UUAS (%) on the Universal Dependencies English Web Treebank (en_ewt). The evaluated models (a) Gemma-3-12B, (b) Qwen-3-8B, and (c) Llama-3.1-8B—achieve peak recoverability $> 60\%$ in the early layers, confirming the pipeline’s efficacy on natural text.

predicts the verb number prescribed by one hypothesis. Figure 14 reports 97.79% for $\mathcal{H}_{syn}$, 98.93% for $\mathcal{H}_{prox}$, and 98.95% for $\mathcal{H}_{maj}$ on the disjoint-vocabulary test set. Thus, all three scalar number features are linearly accessible even though the distance probes do not recover their target graphs. This result does not by itself show that the model uses any feature causally or that the features are geometrically entangled; those questions require the interventions and MAR analysis.

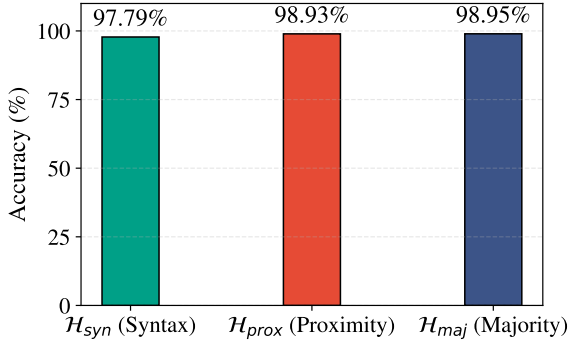


Figure 14: Linear probing accuracies on the final token representations (Layer 45) for Gemma-3-12B using the 4-attractor split. The near-perfect classification scores across all three hypotheses demonstrate that structural, proximity, and majority number features are simultaneously linearly encoded in the residual stream.

C.7 Standard Deviation in Structural Probing Across Random Seeds

Table 5 reports mean and standard deviation of UUAS over seeds 42, 96, and 1120. The low, overlapping results are reproducible across these initial-

izations. Seed stability does not identify why the target graphs are difficult to recover.

D Boundless DAS: Optimization Protocols and Extended Results

D.1 Optimization Strategy and Training Protocol

This section gives the optimization details for the DAS rotation R_θ and mask-boundary parameters (a_θ, c_θ) .

We anneal the sigmoid temperature from 50.0 to 0.1 so that the soft mask approaches a contiguous binary subspace (Wu et al., 2023). We optimize the boundary parameters at learning rate 10^{-2} and the rotation at 10^{-3} , with a 10% warmup.

Epochs and Model-Specific Compute Constraints

Because Boundless DAS requires computing gradients through the full language model during counterfactual interventions, we adjusted our training epochs based on model scale and computational constraints:

- **Gemma-3-12B and Qwen-3-8B:** These models were trained for a maximum of 20 epochs to observe long-term convergence behavior. We implemented early stopping, halting training if the validation-set metric failed to improve by a minimum delta of 0.005 for 3 consecutive epochs.
- **Olmo-3-7B and Llama-3.1-8B:** Due to extensive computational overhead, these mod-

els were trained for exactly 3 epochs without early stopping.

At the end of each epoch, we evaluated IIA on the held-out 1,000-instance validation set. For each layer and hypothesis, we retained the rotation and boundary checkpoint with the best validation IIA.

D.2 Boundless DAS Hyperparameters

Table 6 details the complete optimization configuration used across all experiments.

D.3 Intervention Results on OLMo-3-7B and Llama-3.1-8B

We present the detailed Boundless DAS evaluation for Olmo-3-7B and Llama-3.1-8B in this section. Figure 15 illustrates the layer-wise Interchange Intervention Accuracy (IIA) across all configurations, while Figure 16 demonstrates the effect of systematically varying the number of intermediate attractors on the peak IIA.

OLMo and Llama reach high IIA for several competing hypotheses: $\mathcal{H}_{syn}$, $\mathcal{H}_{prox}$, and $\mathcal{H}_{maj}$ can each steer outputs toward their counterfactual targets. The attractor analysis shows that the pattern varies by model, direction, and hypothesis rather than disappearing uniformly as dependencies lengthen. These results extend the main finding beyond Gemma and Qwen: success for one intervention does not uniquely identify the model’s rule.

D.4 Control Experiment: DAS on a Randomly Initialized Network

Intervention optimizers can fit spurious alignments or memorize their objective (Arora et al., 2024). We therefore test whether the same procedure produces high IIA on an untrained network.

We apply the same DAS protocol to a randomly initialized Qwen-3-8B model on the 4-attractor splits. Because the target is binary, chance IIA is approximately 50%.

Figure 17 presents the layer-wise IIA for this untrained network. As expected, the intervention performance remains uniformly flat at the $\sim 50\%$ random-chance baseline across all network layers. This holds true for both the $SG \rightarrow PL$ and $PL \rightarrow SG$ intervention directions, and across all three causal hypotheses ($\mathcal{H}_{syn}$, $\mathcal{H}_{prox}$, and $\mathcal{H}_{maj}$).

Near-chance IIA in this control shows that the trained-model scores do not follow merely from applying DAS to an untrained network. The result

supports dependence on pretrained parameters, but it does not identify a unique mechanism.

D.5 Standard Deviation in Causal Interventions Across Random Seeds

We run each Boundless DAS experiment with seeds 42, 96, and 1120. Tables 7 and 8 report mean and standard deviation ($\mu \pm \sigma$) for every layer, model, and hypothesis.

The tables show layer- and model-specific variation. We base the main claims on patterns repeated across seeds rather than on a single optimization run.

D.6 Addressing Circularity Risk: Scrambled-Target Control

To test whether DAS relies on lexical correspondence between the base and source sentences, we construct a derangement within each hypothesis split: no base sentence keeps its original source bundle, but the replacement source retains the counterfactual target number. The resulting source is lexically and semantically unrelated to the base while preserving the feature that the intervention is meant to transfer.

Figure 4 shows that IIA remains high for Gemma-3-12B and Qwen-3-8B. The result is consistent with a context-independent number direction rather than memorization of a particular source sentence. It is a partial, not decisive, circularity control: because the scrambled source still carries the correct target number, high IIA does not rule out every way in which an expressive optimizer could exploit that feature. The randomly initialized Qwen control in Figure 17, where IIA remains near chance, provides a separate check that pretrained structure is required in this setup.

E Mechanistic Baseline: Differential Binary Masking (DBM)

We compare rotation-based DAS with Differential Binary Masking (DBM). Boundless DAS learns an orthogonal rotation, whereas DBM acts in the original activation basis. It learns a sparse binary mask over individual coordinates (De Cao et al., 2020; Csordás et al., 2021; Davies et al., 2023; Huang et al., 2024).

E.1 Mathematical Formulation

Let h_b and h_s represent the neural activations at a specific layer for the base and source sentences, respectively. DBM learns a continuous parameter

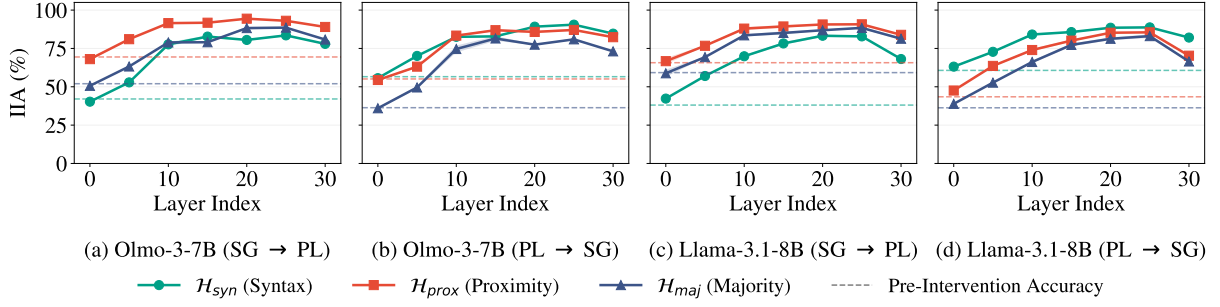


Figure 15: Boundless DAS IIA for OLMo-3-7B and Llama-3.1-8B with four attractors. Dashed lines show hypothesis-specific pre-intervention baselines; shading shows standard deviation over seeds 42, 96, and 1120. Tables 7 and 8 report the numerical values.

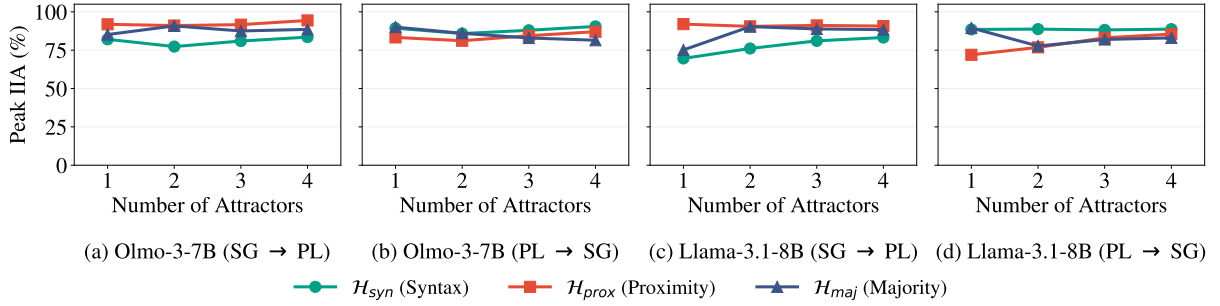


Figure 16: Effect of attractor count on Boundless DAS. We report peak IIA as the number of attractors varies from one to four, separately for each model, intervention direction, and hypothesis. The effect of additional attractors is not uniform: stability depends on the model, direction, and causal hypothesis.

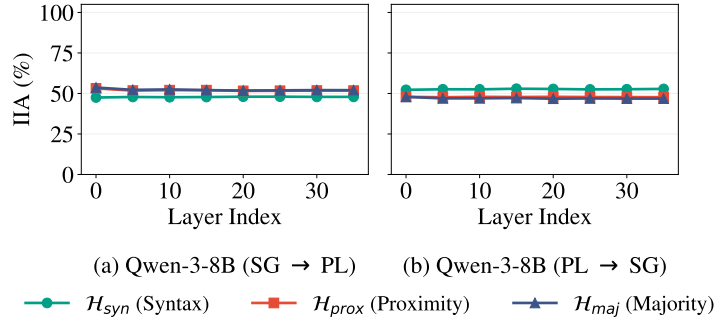


Figure 17: Boundless DAS IIA on a randomly initialized Qwen-3-8B model in the 4-attractor setting. IIA remains near 50% across layers and hypotheses. The same optimizer therefore does not obtain high IIA on this untrained network.

vector $m_\theta \in \mathbb{R}^d$ (where d is the hidden dimension of the layer) that acts as a gate.

To perform the interchange intervention, the base and source representations are mixed using a sigmoid-approximated mask:

$$h_{int} = \left(1 - \sigma\left(\frac{m_\theta}{T}\right)\right) \odot h_b + \sigma\left(\frac{m_\theta}{T}\right) \odot h_s \quad (2)$$

where $\odot$ denotes element-wise multiplication and T is a temperature parameter that controls the steepness of the sigmoid function. As $T \rightarrow 0$, the approximation approaches a hard binary mask, effec-

tively swapping specific individual neurons from the source representation into the base representation.

The objective is to optimize m_θ such that the intervened representation h_{int} forces the language model to output the counterfactual target $y_{\mathcal{H}}$ prescribed by the specific causal hypothesis $\mathcal{H}$. To encourage the model to find the minimal set of neurons responsible for the behavior, an L_1 regularization penalty is applied to the mask. The total

loss is:

$$\mathcal{L}_{DBM} = \text{CE}(\text{LM}(h_{int}), y_{\mathcal{H}}) + \lambda \|m_{\theta}\|_1 \quad (3)$$

where CE is the cross-entropy loss and λ is the regularization coefficient.

E.2 Constructing Intervention Pairs

DBM uses the same hypotheses, base–source pairs, counterfactual targets, and 4-attractor splits as Boundless DAS.

Interventions are evaluated in both the SG $\rightarrow$ PL and PL $\rightarrow$ SG directions. For each configuration:

- **Hierarchical Syntax** ($\mathcal{H}_{syn}$): The source sentence modifies only the morphological number of the main subject relative to the base sentence.
- **Proximity Recency** ($\mathcal{H}_{prox}$): The source sentence maintains the base main subject, but flips the number of the linearly closest attractor noun.
- **Majority Number** ($\mathcal{H}_{maj}$): The source sentence flips the aggregate majority mode of the nouns within the context window.

E.3 Hyperparameters and Training Details

Consistent with the training protocol established in the RAVEL benchmark (Huang et al., 2024), the temperature parameter T was systematically annealed during training to ensure the soft mask converged into a strict binary selection.

Due to the element-wise nature of the mask optimization, DBM converges faster than orthogonal rotation searches. Consequently, DBM was trained for exactly 3 epochs across all evaluated models (Gemma-3-12B, Qwen-3-8B, Llama-3.1-8B, and Olmo-3-7B). Table 9 details the complete optimization configuration.

| Hyperparameter | Value |
|------------------------------------|----------------------|
| Optimizer | AdamW |
| Precision | bfloat16 |
| Epochs | 3 |
| Learning Rate | 1.0×10^{-4} |
| L_1 Regularization (λ) | 0.001 |
| Initial Mask Temp (T_{start}) | 1.0×10^{-2} |
| Final Mask Temp (T_{end}) | 1.0×10^{-7} |
| Random Seeds | 42, 96, 1120 |

Table 9: Training hyperparameters for Differential Binary Masking (DBM) optimization.

Similar to the DAS evaluation protocol, the models were evaluated on the held-out 1,000-instance validation set at the end of every epoch. Only the best-performing mask checkpoint evaluated on the validation set for each layer and hypothesis was retained for the final evaluations on test set.

E.4 Intervention Results and Analysis

The Gemma-3-12B and Qwen-3-8B layer-wise curves are shown beside DAS in Figure 2; Figure 19 gives Llama-3.1-8B and OLMo-3-7B. DBM remains close to the pre-intervention baselines across hypotheses and models, and for OLMo-3-7B in SG $\rightarrow$ PL it falls below baseline. Figures 18 and 20 show that this pattern persists as attractor count changes. In contrast, the rotated DAS search succeeds in several of the same settings. This supports the limited conclusion that sparse masking in the original basis does not recover the effective intervention directions found after rotation; it does not prove that the model has a unique distributed circuit.

F Manifold Alignment Ratio (MAR): Mathematical Proof and Extended Results

F.1 Mathematical Proof of the MAR

We provide the formal proof that the Manifold Alignment Ratio (MAR) defined in Equation 1 corresponds precisely to the scaled sum of the squared cosines of the principal angles between two vector subspaces, ensuring it acts as a rotation-invariant measure of geometric intersection even when the subspace dimensions differ.

Let $\mathcal{U}$ and $\mathcal{V}$ be two subspaces within an ambient inner product space $\mathbb{R}^d$ with dimensions m_u and m_v , respectively. These are spanned by the orthonormal columns of matrices $W_U \in \mathbb{R}^{d \times m_u}$ and $W_V \in \mathbb{R}^{d \times m_v}$. The number of principal angles between $\mathcal{U}$ and $\mathcal{V}$ is bounded by $k = \min(m_u, m_v)$. These angles $\theta_1, \theta_2, \dots, \theta_k \in [0, \pi/2]$ are defined such that their cosines correspond to the singular values of the cross-basis inner product matrix:

$$\text{SVD}(W_U^\top W_V) = U_\Sigma \Sigma V_\Sigma^\top$$

where Σ contains the singular values $\cos \theta_1, \cos \theta_2, \dots, \cos \theta_k$ along its diagonal.

The orthogonal projection operators onto $\mathcal{U}$ and $\mathcal{V}$ are given by $P_U = W_U W_U^\top$ and $P_V = W_V W_V^\top$. To evaluate the numerator of the MAR, we compute

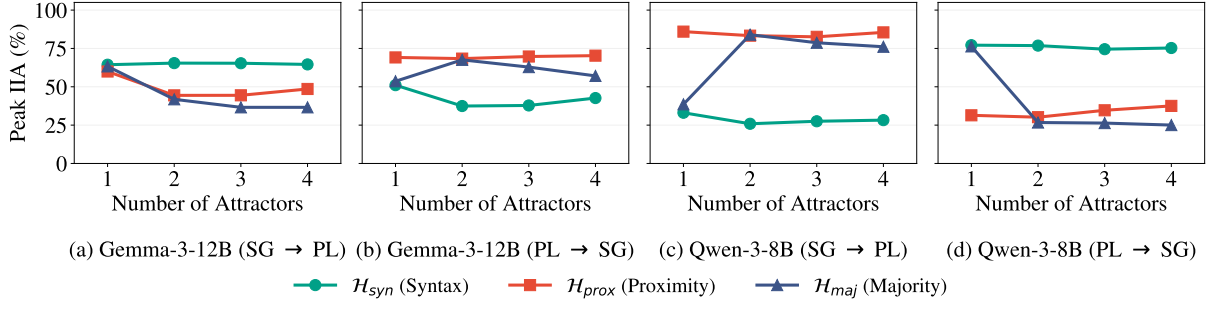


Figure 18: Peak DBM IIA as a function of attractor count (1–4) for Gemma-3-12B and Qwen-3-8B. Across both models and directions, IIA remains low. See Figure 20 for Llama-3.1-8B and OLMo-3-7B.

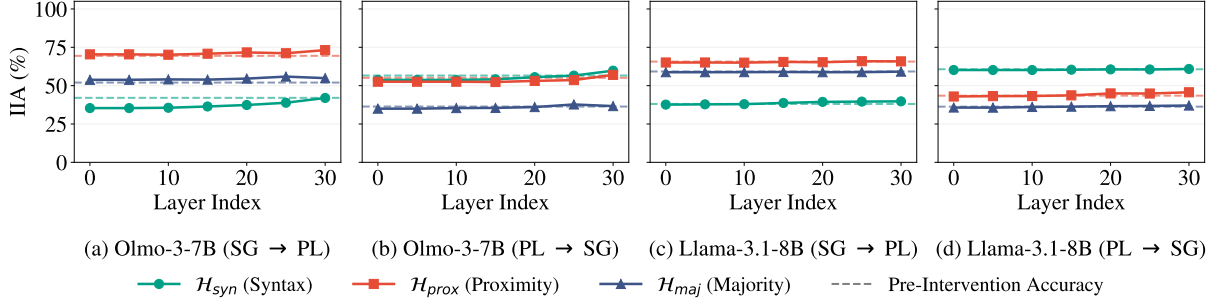


Figure 19: Layer-wise DBM IIA for OLMo-3-7B and Llama-3.1-8B with four attractors. Curves remain near the counterfactual baselines; OLMo-3-7B falls below baseline in SG→PL.

the trace of the product of these projection operators:

$$\begin{aligned}
 \text{Tr}(P_U P_V) &= \text{Tr}(W_U W_U^\top W_V W_V^\top) \\
 &= \text{Tr}(W_U^\top W_V W_V^\top W_U) \\
 &\quad \text{(by cyclic invariance)} \\
 &= \text{Tr}\left((W_U^\top W_V)(W_U^\top W_V)^\top\right)
 \end{aligned}$$

Substituting the singular value decomposition of $W_U^\top W_V$ into this expression yields:

$$\begin{aligned}
 \text{Tr}(P_U P_V) &= \text{Tr}(U_\Sigma \Sigma V_\Sigma^\top V_\Sigma \Sigma^\top U_\Sigma^\top) \\
 &= \text{Tr}(U_\Sigma \Sigma \Sigma^\top U_\Sigma^\top) \\
 &= \text{Tr}(\Sigma \Sigma^\top) = \sum_{i=1}^k \cos^2 \theta_i
 \end{aligned}$$

The denominator of the MAR relies on the trace of the individual projection operators. Because the trace of a projection matrix equals the rank of its subspace, $\text{Tr}(P_U) = m_u$ and $\text{Tr}(P_V) = m_v$. Thus, the MAR simplifies to:

$$\text{MAR}(P_U, P_V) = \frac{1}{\sqrt{m_u m_v}} \sum_{i=1}^k \cos^2 \theta_i$$

Thus, MAR is the sum of squared cosines of the principal angles, normalized by the geometric mean of

the subspace dimensions. Because principal angles are invariant to orthonormal basis choice, rotations within either subspace do not change MAR.

F.2 Extended Geometric Analysis for Llama-3.1-8B and OLMo-3-7B

In (§8), we introduced the Manifold Alignment Ratio (MAR) to quantify the geometric overlap between the learned causal subspaces of conflicting hypotheses. While our primary analysis detailed the geometric properties of Gemma-3-12B and Qwen-3-8B, we present the analogous subspace dimension analysis and MAR scores for Llama-3.1-8B and OLMo-3-7B in this section.

Subspace Dimensionality and Singular Bias.

Figure 21 illustrates the dimensions of the learned intervention subspaces for Llama-3.1-8B and OLMo-3-7B. Extending the findings from Figure 5, we evaluate whether these models exhibit a similar geometric asymmetry when steering predictions toward singular versus plural targets.

As noted in §7, plural-target interventions often use larger subspaces than singular-target interventions.

MAR Values. Table 10 reports the selected-run MAR values for all four models, while Table 11

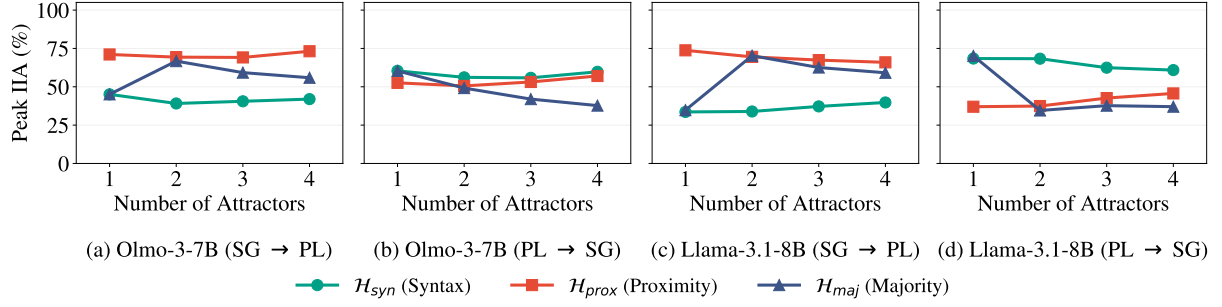


Figure 20: Peak DBM IIA by attractor count for OLmo-3-7B and Llama-3.1-8B. The low-alignment pattern persists as the prompt becomes longer.

reports mean and standard deviation across seeds. The main text focuses on Gemma-3-12B and Qwen-3-8B; the tables here provide the corresponding Llama-3.1-8B and OLmo-3-7B results and a consolidated view.

| Model | Direction | MAR_{prox} | MAR_{maj} |
|--------------|---------------------|--------------|-------------|
| Gemma-3-12B | SG $\rightarrow$ PL | 0.307 | 0.273 |
| | PL $\rightarrow$ SG | 0.593 | 0.356 |
| Qwen-3-8B | SG $\rightarrow$ PL | 0.165 | 0.139 |
| | PL $\rightarrow$ SG | 0.114 | 0.161 |
| Llama-3.1-8B | SG $\rightarrow$ PL | 0.378 | 0.385 |
| | PL $\rightarrow$ SG | 0.422 | 0.400 |
| OLmo-3-7B | SG $\rightarrow$ PL | 0.550 | 0.470 |
| | PL $\rightarrow$ SG | 0.548 | 0.538 |

Table 10: Selected-run MAR values at the optimal 4-attractor intervention layers. Higher values indicate greater overlap between the syntax subspace and the corresponding heuristic subspace.

| Model | Direction | MAR_{prox} | MAR_{maj} |
|--------------|---------------------|---------------------|---------------------|
| Gemma-3-12B | PL $\rightarrow$ SG | 0.4608 ± 0.1596 | 0.5570 ± 0.1968 |
| | SG $\rightarrow$ PL | 0.3907 ± 0.0783 | 0.3393 ± 0.0685 |
| Qwen-3-8B | PL $\rightarrow$ SG | 0.3574 ± 0.2221 | 0.3373 ± 0.2253 |
| | SG $\rightarrow$ PL | 0.2571 ± 0.0819 | 0.4281 ± 0.2505 |
| Llama-3.1-8B | PL $\rightarrow$ SG | 0.5817 ± 0.1386 | 0.5729 ± 0.1505 |
| | SG $\rightarrow$ PL | 0.5222 ± 0.1251 | 0.5524 ± 0.1452 |
| OLmo-3-7B | PL $\rightarrow$ SG | 0.7453 ± 0.1709 | 0.7213 ± 0.1588 |
| | SG $\rightarrow$ PL | 0.7262 ± 0.1545 | 0.6801 ± 0.1821 |

Table 11: MAR mean $\pm$ standard deviation across seeds 42, 96, and 1120 for all evaluated models. Similar IIA can coexist with appreciable variation in cross-hypothesis subspace geometry.

F.3 Statistical Significance of MAR: A Monte Carlo Null Baseline

We use a Monte Carlo random-subspace null to test whether each observed MAR exceeds the overlap expected from its dimensions.

For each model and intervention configuration, given the ambient hidden dimension d and the spe-

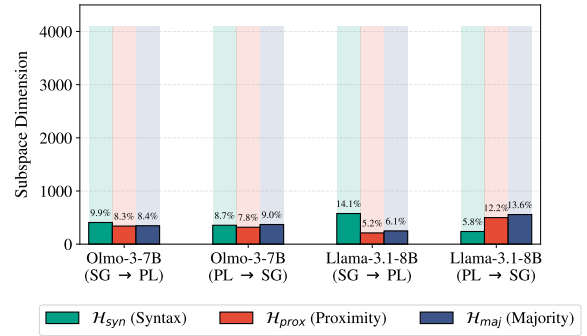


Figure 21: Learned causal subspace dimensionality for the optimal intervention layers in Llama-3.1-8B and OLmo-3-7B (4-attractor setup). The percentages annotated above the bars indicate the relative size of the subspace compared to the model’s total hidden dimension (represented by the shaded region: 4096 for both Llama-3.1-8B and OLmo-3-7B). The geometric asymmetry reveals differences in the representational space required to steer the model toward plural versus singular verbs, complementing the structural bias observations detailed in §7.

cific dimensions of our learned causal subspaces m_{syn} and m_{heur} , we executed 1,000 iterations of the following procedure:

- We sampled two random Gaussian matrices of shape (d, m_{syn}) and (d, m_{heur}) .
- We orthogonalized these matrices via QR decomposition to yield valid orthonormal bases, Q_{syn} and Q_{heur} , representing uniformly distributed random subspaces.
- We computed the random MAR baseline using the Frobenius norm formulation:

$$\text{MAR} = \frac{\|Q_{syn}^T Q_{heur}\|_F^2}{\sqrt{m_{syn} \cdot m_{heur}}}$$

- Finally, we computed the Z-score to measure how many standard deviations our empirically observed MAR (MAR_{obs}) deviated from the random null mean (μ_{null}):

$$Z = \frac{\text{MAR}_{obs} - \mu_{null}}{\sigma_{null}}$$

Table 12 reports the dimension-matched results. Null means range from about 0.07 to 0.14; the evaluated learned subspaces have larger MAR values, with reported normal-approximation Z-scores from 79.93 to above 1,500 and right-tailed $p < 10^{-10}$. This rules out ordinary random-subspace intersection for these runs. It does not show that either hypothesis is used uniquely during unmodified generation.

F.4 Standard Deviation in Geometric Overlap and Representational Redundancy

Tables 1 and 11 report mean and standard deviation over three DAS initializations. IIA is comparatively stable (Tables 7 and 8), while MAR varies more. Thus, different optimization runs can find similarly effective interventions with different cross-hypothesis geometry. This may reflect representational redundancy or optimizer non-identifiability; the present experiments do not distinguish those explanations.

| Hypothesis | Sentence ($N = 3$ Attractors) | Data Split |
|-------------------------------------|---|---------------------------------|
| <i>Panel A: Animate Sentences</i> | | |
| Syntax ($\mathcal{H}_{syn}$) | The athlete that the coach trains whom the manager directs that the sponsors fund wins . | $\mathcal{H}_{prox}$ (SG Setup) |
| | The athletes that the coaches train whom the managers direct that the sponsor funds win . | $\mathcal{H}_{prox}$ (PL Setup) |
| | The patient whom the doctors treat that the nurses comfort who the visitor likes sleeps . | $\mathcal{H}_{maj}$ (SG Setup) |
| | The patients whom the doctor treats that the nurse comforts who the visitors like sleep . | $\mathcal{H}_{maj}$ (PL Setup) |
| Proximity ($\mathcal{H}_{prox}$) | The athlete that the coach trains whom the manager directs that the sponsors fund win . | $\mathcal{H}_{prox}$ (SG Setup) |
| | The athletes that the coaches train whom the managers direct that the sponsor funds wins . | $\mathcal{H}_{prox}$ (PL Setup) |
| Majority ($\mathcal{H}_{maj}$) | The patient whom the doctors treat that the nurses comfort who the visitor likes sleep . | $\mathcal{H}_{maj}$ (SG Setup) |
| | The patients whom the doctor treats that the nurse comforts who the visitors like sleeps . | $\mathcal{H}_{maj}$ (PL Setup) |
| <i>Panel B: Inanimate Sentences</i> | | |
| Syntax ($\mathcal{H}_{syn}$) | The movie that the guard watches that the critic reviews that the sponsors fund ends . | $\mathcal{H}_{prox}$ (SG Setup) |
| | The movies that the guards watch that the critics review that the sponsor funds end . | $\mathcal{H}_{prox}$ (PL Setup) |
| | The car that the mechanics fix that the drivers buy that the thief steals breaks . | $\mathcal{H}_{maj}$ (SG Setup) |
| | The cars that the mechanic fixes that the driver buys that the thieves steal break . | $\mathcal{H}_{maj}$ (PL Setup) |
| Proximity ($\mathcal{H}_{prox}$) | The movie that the guard watches that the critic reviews that the sponsors fund end . | $\mathcal{H}_{prox}$ (SG Setup) |
| | The movies that the guards watch that the critics review that the sponsor funds ends . | $\mathcal{H}_{prox}$ (PL Setup) |
| Majority ($\mathcal{H}_{maj}$) | The car that the mechanics fix that the drivers buy that the thief steals break . | $\mathcal{H}_{maj}$ (SG Setup) |
| | The cars that the mechanic fixes that the driver buys that the thieves steal breaks . | $\mathcal{H}_{maj}$ (PL Setup) |

Table 3: A representative sample of 16 syntactically complex sentences (8 animate, 8 inanimate) from the SYNTHAX dataset ($N = 3$). For each subset, the target verb is actively flipped to reflect the counterfactual prediction of the corresponding causal hypothesis. The specific nominal features dictating that prediction are highlighted accordingly.

| Hyperparameter | Value |
|---------------------------------|--------------|
| Optimizer | Adam |
| Initial Learning Rate | 0.001 |
| Momentum (β_1, β_2) | (0.9, 0.999) |
| Epsilon (ϵ) | 10^{-8} |
| Batch Size | 20 |
| Max Epochs | 30 |
| LR Decay Factor | 0.1 |
| Random Seeds | 42, 96, 1120 |

Table 4: Hyperparameters for Structural Probing across models.

| Layer | $\mathcal{H}_{syn}$ | $\mathcal{H}_{prox}$ | $\mathcal{H}_{maj}$ |
|---------------------|---------------------|----------------------|---------------------|
| Gemma-3-12B | | | |
| 0 | 0.0625 ± 0.0007 | 0.0611 ± 0.0009 | 0.0632 ± 0.0014 |
| 5 | 0.0524 ± 0.0003 | 0.0514 ± 0.0001 | 0.0543 ± 0.0010 |
| 10 | 0.0454 ± 0.0001 | 0.0464 ± 0.0018 | 0.0441 ± 0.0020 |
| 15 | 0.0940 ± 0.0001 | 0.0956 ± 0.0003 | 0.0771 ± 0.0281 |
| 20 | 0.1463 ± 0.0004 | 0.1457 ± 0.0005 | 0.1463 ± 0.0006 |
| 25 | 0.0696 ± 0.0013 | 0.0697 ± 0.0010 | 0.0722 ± 0.0010 |
| 30 | 0.0386 ± 0.0009 | 0.0413 ± 0.0014 | 0.0420 ± 0.0020 |
| 35 | 0.0297 ± 0.0006 | 0.0324 ± 0.0010 | 0.0310 ± 0.0017 |
| 40 | 0.0350 ± 0.0002 | 0.0328 ± 0.0022 | 0.0362 ± 0.0021 |
| 45 | 0.0602 ± 0.0004 | 0.0515 ± 0.0013 | 0.0587 ± 0.0011 |
| Qwen-3-8B | | | |
| 0 | 0.0495 ± 0.0011 | 0.0522 ± 0.0025 | 0.0453 ± 0.0027 |
| 5 | 0.0860 ± 0.0087 | 0.0857 ± 0.0086 | 0.0886 ± 0.0121 |
| 10 | 0.0460 ± 0.0006 | 0.0449 ± 0.0002 | 0.0460 ± 0.0003 |
| 15 | 0.0614 ± 0.0005 | 0.0627 ± 0.0002 | 0.0610 ± 0.0010 |
| 20 | 0.0766 ± 0.0004 | 0.0779 ± 0.0005 | 0.0765 ± 0.0006 |
| 25 | 0.0673 ± 0.0003 | 0.0665 ± 0.0002 | 0.0693 ± 0.0012 |
| 30 | 0.0951 ± 0.0199 | 0.1023 ± 0.0235 | 0.1004 ± 0.0194 |
| 35 | 0.0783 ± 0.0187 | 0.0868 ± 0.0121 | 0.0705 ± 0.0206 |
| Llama-3.1-8B | | | |
| 0 | 0.1954 ± 0.0036 | 0.1953 ± 0.0034 | 0.1910 ± 0.0057 |
| 5 | 0.1858 ± 0.0459 | 0.1857 ± 0.0357 | 0.1872 ± 0.0502 |
| 10 | 0.3338 ± 0.0493 | 0.3236 ± 0.0486 | 0.3577 ± 0.0727 |
| 15 | 0.2229 ± 0.0463 | 0.2162 ± 0.0535 | 0.2363 ± 0.0549 |
| 20 | 0.1248 ± 0.0099 | 0.1143 ± 0.0008 | 0.1167 ± 0.0047 |
| 25 | 0.1277 ± 0.0195 | 0.1484 ± 0.0214 | 0.1339 ± 0.0353 |
| 30 | 0.0611 ± 0.0132 | 0.0674 ± 0.0326 | 0.0528 ± 0.0006 |

Table 5: Structural probing Undirected Unlabeled Attachment Score (UUAS) represented as Mean $\pm$ Standard Deviation ($\mu \pm \sigma$) across three random seeds (42, 96, 1120).

| Hyperparameter | Value |
|---------------------------------------|----------------------|
| Optimizer | AdamW |
| Precision | bfloat16 |
| Rotation Learning Rate | 1.0×10^{-3} |
| Boundary Learning Rate | 1.0×10^{-2} |
| Warmup Ratio | 0.1 |
| Initial Mask Temp (β_{start}) | 50.0 |
| Final Mask Temp (β_{end}) | 0.1 |
| Boundary Loss Weight | 1.0 |
| Early Stopping Patience | 3 epochs |
| Early Stopping Min Delta | 0.005 |
| Random Seeds | 42, 96, 1120 |

Table 6: Training hyperparameters for Boundless DAS optimization.

| Layer | $\mathcal{H}_{syn}$ | $\mathcal{H}_{prox}$ | $\mathcal{H}_{maj}$ |
|---|---------------------|----------------------|---------------------|
| Gemma-3-12B ($SG \rightarrow PL$) | | | |
| 0 | 0.7890 ± 0.0189 | 0.6857 ± 0.0263 | 0.5649 ± 0.0395 |
| 5 | 0.9253 ± 0.0186 | 0.8842 ± 0.0025 | 0.8148 ± 0.0271 |
| 10 | 0.9440 ± 0.0055 | 0.9070 ± 0.0049 | 0.8202 ± 0.0213 |
| 15 | 0.9244 ± 0.0048 | 0.8317 ± 0.0065 | 0.7622 ± 0.0053 |
| 20 | 0.9488 ± 0.0049 | 0.8340 ± 0.0061 | 0.7749 ± 0.0223 |
| 25 | 0.9526 ± 0.0035 | 0.8977 ± 0.0177 | 0.8406 ± 0.0364 |
| 30 | 0.9737 ± 0.0058 | 0.9002 ± 0.0304 | 0.8529 ± 0.0350 |
| 35 | 0.9738 ± 0.0006 | 0.8941 ± 0.0236 | 0.8694 ± 0.0138 |
| 40 | 0.9656 ± 0.0061 | 0.9098 ± 0.0089 | 0.8691 ± 0.0397 |
| 45 | 0.9856 ± 0.0029 | 0.9467 ± 0.0066 | 0.9320 ± 0.0058 |
| Gemma-3-12B ($PL \rightarrow SG$) | | | |
| 0 | 0.6500 ± 0.0130 | 0.8458 ± 0.0299 | 0.7474 ± 0.0251 |
| 5 | 0.7762 ± 0.0359 | 0.9776 ± 0.0036 | 0.9457 ± 0.0147 |
| 10 | 0.8289 ± 0.0190 | 0.9528 ± 0.0166 | 0.9194 ± 0.0037 |
| 15 | 0.8264 ± 0.0057 | 0.9419 ± 0.0070 | 0.8877 ± 0.0224 |
| 20 | 0.8302 ± 0.0160 | 0.9542 ± 0.0071 | 0.9364 ± 0.0068 |
| 25 | 0.8842 ± 0.0055 | 0.9636 ± 0.0049 | 0.9441 ± 0.0018 |
| 30 | 0.8840 ± 0.0262 | 0.9731 ± 0.0037 | 0.9673 ± 0.0057 |
| 35 | 0.9048 ± 0.0063 | 0.9746 ± 0.0022 | 0.9681 ± 0.0052 |
| 40 | 0.8906 ± 0.0252 | 0.9712 ± 0.0095 | 0.9507 ± 0.0038 |
| 45 | 0.9323 ± 0.0084 | 0.9868 ± 0.0017 | 0.9862 ± 0.0016 |
| Qwen-3-8B ($SG \rightarrow PL$) | | | |
| 0 | 0.1974 ± 0.0011 | 0.8536 ± 0.0018 | 0.7690 ± 0.0019 |
| 5 | 0.2490 ± 0.0029 | 0.8593 ± 0.0020 | 0.7772 ± 0.0031 |
| 10 | 0.3602 ± 0.0061 | 0.8641 ± 0.0027 | 0.7934 ± 0.0077 |
| 15 | 0.5355 ± 0.0038 | 0.8661 ± 0.0037 | 0.7944 ± 0.0052 |
| 20 | 0.6298 ± 0.0077 | 0.9100 ± 0.0053 | 0.8447 ± 0.0104 |
| 25 | 0.5459 ± 0.0049 | 0.9273 ± 0.0018 | 0.8619 ± 0.0120 |
| 30 | 0.5739 ± 0.0063 | 0.9338 ± 0.0072 | 0.8756 ± 0.0113 |
| 35 | 0.4389 ± 0.0020 | 0.8961 ± 0.0055 | 0.8279 ± 0.0091 |
| Qwen-3-8B ($PL \rightarrow SG$) | | | |
| 0 | 0.7220 ± 0.0011 | 0.3151 ± 0.0010 | 0.2369 ± 0.0011 |
| 5 | 0.7313 ± 0.0022 | 0.3427 ± 0.0044 | 0.2846 ± 0.0036 |
| 10 | 0.7644 ± 0.0020 | 0.4366 ± 0.0099 | 0.3601 ± 0.0097 |
| 15 | 0.7830 ± 0.0056 | 0.6063 ± 0.0111 | 0.5271 ± 0.0045 |
| 20 | 0.8445 ± 0.0069 | 0.6958 ± 0.0052 | 0.5939 ± 0.0038 |
| 25 | 0.8611 ± 0.0021 | 0.6500 ± 0.0027 | 0.5070 ± 0.0134 |
| 30 | 0.8796 ± 0.0080 | 0.6930 ± 0.0096 | 0.5563 ± 0.0076 |
| 35 | 0.8594 ± 0.0037 | 0.5479 ± 0.0060 | 0.3422 ± 0.0060 |

Table 7: Interchange Intervention Accuracy (IIA) scores represented as Mean $\pm$ Standard Deviation ($\mu \pm \sigma$) across three different random seeds (42, 96, 1120) for Gemma-3-12B and Qwen-3-8B (4-attractor configuration).

| Layer | $\mathcal{H}_{syn}$ | $\mathcal{H}_{prox}$ | $\mathcal{H}_{maj}$ |
|--|---------------------|----------------------|---------------------|
| Llama-3.1-8B ($SG \rightarrow PL$) | | | |
| 0 | 0.4247 ± 0.0015 | 0.6884 ± 0.0212 | 0.6147 ± 0.0238 |
| 5 | 0.5699 ± 0.0018 | 0.7646 ± 0.0025 | 0.6904 ± 0.0026 |
| 10 | 0.6918 ± 0.0062 | 0.8680 ± 0.0098 | 0.8273 ± 0.0070 |
| 15 | 0.7728 ± 0.0084 | 0.8833 ± 0.0093 | 0.8387 ± 0.0107 |
| 20 | 0.8254 ± 0.0090 | 0.9054 ± 0.0018 | 0.8658 ± 0.0025 |
| 25 | 0.8254 ± 0.0033 | 0.9033 ± 0.0037 | 0.8779 ± 0.0067 |
| 30 | 0.6806 ± 0.0032 | 0.8348 ± 0.0040 | 0.8060 ± 0.0087 |
| Llama-3.1-8B ($PL \rightarrow SG$) | | | |
| 0 | 0.6381 ± 0.0060 | 0.4740 ± 0.0017 | 0.3903 ± 0.0015 |
| 5 | 0.7220 ± 0.0056 | 0.6310 ± 0.0047 | 0.5334 ± 0.0061 |
| 10 | 0.8339 ± 0.0064 | 0.7302 ± 0.0082 | 0.6556 ± 0.0081 |
| 15 | 0.8514 ± 0.0048 | 0.7923 ± 0.0076 | 0.7621 ± 0.0098 |
| 20 | 0.8812 ± 0.0034 | 0.8529 ± 0.0010 | 0.8080 ± 0.0090 |
| 25 | 0.8862 ± 0.0030 | 0.8506 ± 0.0053 | 0.8168 ± 0.0118 |
| 30 | 0.8172 ± 0.0032 | 0.7022 ± 0.0050 | 0.6633 ± 0.0047 |
| OLMo-3-7B ($SG \rightarrow PL$) | | | |
| 0 | 0.4108 ± 0.0069 | 0.6916 ± 0.0098 | 0.5100 ± 0.0034 |
| 5 | 0.5260 ± 0.0023 | 0.8101 ± 0.0013 | 0.6312 ± 0.0025 |
| 10 | 0.7612 ± 0.0137 | 0.9120 ± 0.0026 | 0.7829 ± 0.0056 |
| 15 | 0.8183 ± 0.0079 | 0.9160 ± 0.0029 | 0.7860 ± 0.0045 |
| 20 | 0.8004 ± 0.0046 | 0.9431 ± 0.0008 | 0.8730 ± 0.0087 |
| 25 | 0.8278 ± 0.0064 | 0.9333 ± 0.0026 | 0.8796 ± 0.0054 |
| 30 | 0.7807 ± 0.0031 | 0.8869 ± 0.0038 | 0.8080 ± 0.0013 |
| OLMo-3-7B ($PL \rightarrow SG$) | | | |
| 0 | 0.5637 ± 0.0065 | 0.5530 ± 0.0077 | 0.3658 ± 0.0050 |
| 5 | 0.6977 ± 0.0030 | 0.6260 ± 0.0048 | 0.4852 ± 0.0091 |
| 10 | 0.8208 ± 0.0044 | 0.8287 ± 0.0062 | 0.7230 ± 0.0225 |
| 15 | 0.8243 ± 0.0044 | 0.8627 ± 0.0059 | 0.7989 ± 0.0154 |
| 20 | 0.8874 ± 0.0034 | 0.8524 ± 0.0074 | 0.7738 ± 0.0034 |
| 25 | 0.9013 ± 0.0034 | 0.8638 ± 0.0060 | 0.8043 ± 0.0058 |
| 30 | 0.8463 ± 0.0044 | 0.8222 ± 0.0041 | 0.7300 ± 0.0023 |

Table 8: Interchange Intervention Accuracy (IIA) scores represented as Mean $\pm$ Standard Deviation ($\mu \pm \sigma$) across three different random seeds (42, 96, 1120) for Llama-3.1-8B and OLMo-3-7B (4-attractor configuration).

| Model | Direction | Comparison | Null Mean $\pm$ Std | Z-score | P-value |
|--------------|---------------------|----------------------|---------------------|---------|--------------|
| Gemma-3-12B | PL $\rightarrow$ SG | $\mathcal{H}_{prox}$ | 0.0702 ± 0.0003 | 1545.28 | $< 10^{-10}$ |
| Gemma-3-12B | PL $\rightarrow$ SG | $\mathcal{H}_{maj}$ | 0.1419 ± 0.0003 | 753.18 | $< 10^{-10}$ |
| Gemma-3-12B | SG $\rightarrow$ PL | $\mathcal{H}_{prox}$ | 0.1211 ± 0.0003 | 637.50 | $< 10^{-10}$ |
| Gemma-3-12B | SG $\rightarrow$ PL | $\mathcal{H}_{maj}$ | 0.1186 ± 0.0003 | 518.19 | $< 10^{-10}$ |
| Qwen-3-8B | PL $\rightarrow$ SG | $\mathcal{H}_{prox}$ | 0.0895 ± 0.0003 | 79.93 | $< 10^{-10}$ |
| Qwen-3-8B | PL $\rightarrow$ SG | $\mathcal{H}_{maj}$ | 0.1113 ± 0.0003 | 163.54 | $< 10^{-10}$ |
| Qwen-3-8B | SG $\rightarrow$ PL | $\mathcal{H}_{prox}$ | 0.1089 ± 0.0003 | 185.67 | $< 10^{-10}$ |
| Qwen-3-8B | SG $\rightarrow$ PL | $\mathcal{H}_{maj}$ | 0.1056 ± 0.0003 | 109.83 | $< 10^{-10}$ |
| Llama-3.1-8B | PL $\rightarrow$ SG | $\mathcal{H}_{prox}$ | 0.0842 ± 0.0003 | 1102.64 | $< 10^{-10}$ |
| Llama-3.1-8B | PL $\rightarrow$ SG | $\mathcal{H}_{maj}$ | 0.0891 ± 0.0003 | 1055.33 | $< 10^{-10}$ |
| Llama-3.1-8B | SG $\rightarrow$ PL | $\mathcal{H}_{prox}$ | 0.0826 ± 0.0003 | 969.59 | $< 10^{-10}$ |
| Llama-3.1-8B | SG $\rightarrow$ PL | $\mathcal{H}_{maj}$ | 0.0957 ± 0.0003 | 933.25 | $< 10^{-10}$ |
| OLMo-3-7B | PL $\rightarrow$ SG | $\mathcal{H}_{prox}$ | 0.0834 ± 0.0003 | 1466.05 | $< 10^{-10}$ |
| OLMo-3-7B | PL $\rightarrow$ SG | $\mathcal{H}_{maj}$ | 0.0874 ± 0.0003 | 1355.08 | $< 10^{-10}$ |
| OLMo-3-7B | SG $\rightarrow$ PL | $\mathcal{H}_{prox}$ | 0.0854 ± 0.0003 | 1519.83 | $< 10^{-10}$ |
| OLMo-3-7B | SG $\rightarrow$ PL | $\mathcal{H}_{maj}$ | 0.0972 ± 0.0003 | 1185.99 | $< 10^{-10}$ |

Table 12: Monte Carlo simulation results establishing the random-subspace null distribution for MAR across all evaluated models. P-values reported as $< 10^{-10}$ represent cases where the empirically observed MAR is so many standard deviations away from the null mean that the probability of occurrence by random chance is effectively zero.